Firm Scope and Innovation: The Role of Intangibles*

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Abstract

This paper develops an endogenous growth model that integrates vertical and horizontal innovation within a framework of intangible heterogeneity. In the model, I distinguish between transferable intangibles (e.g., R&D, patents) and those embedded within the firm (e.g., brand value, organizational capital). This distinction provides a micro-founded foundation for analyzing firms' investment decisions, their choice of innovation direction, and the resulting aggregate consequences. To discipline and validate the model, I use Compustat segment-level data to construct intangible composition through scope and document a systematic shift in firms' investment patterns as they expand their scope. In the quantitative analysis, I demonstrate that the composition of intangibles within firms is crucial for shaping innovation direction. The results show that conventional size-based policies are a blunt instrument: they fail to account for firm heterogeneity and risk penalizing the very firms that generate the largest knowledge spillovers. In contrast, I find that scope-dependent policies based on intangible investment externalities are more effective than those based on firm size.

Keywords: Schumpeterian growth, step-by-step innovation, intangibles, firm dynamics, span of control.

JEL Classification: E22, O31, O32, O33, O34.

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1. Introduction

Horizontal and vertical innovation are two core strategies in the modern endogenous growth literature for generating long-run growth through creative destruction. Firms can engage in horizontal innovation by expanding their product portfolios (Klette and Kortum, 2004) or pursue vertical innovation by improving the quality of existing products (Aghion, Harris, Howitt, and Vickers, 2001). In parallel, the literature shows that firms invest in intangibles not only to innovate but also to build intangible assets such as brand value and organizational capital that strengthen market positions rather than drive innovation-led growth (Cavenaile and Roldan-Blanco, 2021; Crouzet and Eberly, 2019).

While the directions of innovation and the heterogeneity of intangibles are well documented, their interaction remains largely unexplored. For instance, Apple is a highly vertically integrated firm focused primarily on mobile communication technologies, with a ratio of R&D to firm-specific intangible investment of 2.96. In contrast, 3M is a horizontally integrated company active in the transportation, safety, consumer, and health care industries, with a corresponding ratio of 1.21. One possible explanation is that firms with large stocks of patents and R&D within a sector are more likely to pursue vertical innovation, whereas firms with greater brand value and organizational capital tend to favor horizontal innovation. Conversely, a firm's innovation direction can shape its intangible investments: vertically oriented firms allocate more resources to technology-related intangibles to secure technological superiority and market share, while horizontally oriented firms invest in brand and organizational capital to exploit economies of scope.

This paper fills that gap by answering two questions: (i) why do firms choose different innovation paths under intangible heterogeneity, and (ii) how does the joint determination of innovation direction and intangible investment affect economic growth, creative destruction, and welfare? First, I show that intangible heterogeneity shapes firms' innovation choices: even firms of similar size prefer different innovation paths depending on their types of intangible endowments. Second, I argue that not all intangibles generate the same externalities; different types of intangibles give rise to different sets of exter-

nalities. I find that size-based policies fail to capture firms' investment behavior and can penalize firms that generate large knowledge spillovers. By contrast, scope-dependent policies that target the externalities associated with intangible heterogeneity are more effective than size-based interventions.

To formalize intangible heterogeneity, I classify intangible assets according to their spillover effects across firms. In this paper, transferable intangibles consist of patents and software [1], along with the associated R&D investments. These assets can be transferred between firms and generate spillover effects due to their non-rivalrous and partially excludable nature (Romer, 1990; Aghion and Howitt, 1992; Grossman and Helpman, 1991). In contrast, embedded intangibles comprise brand value and organizational capital are inherently firm-specific and inseparable from the firm that created them. They primarily provide a firm-specific comparative advantage and do not generate spillovers. Consequently, when a firm exits the market, the economic value of embedded intangibles becomes a sunk cost.

I further subdivide embedded intangibles based on their effects on demand and supply side (Table 1). Brand value [2] is a demand shifter, positively influencing the perceived quality of a firm's output (Cavenaile and Roldan-Blanco, 2021; Cavenaile, Celik, Roldan-Blanco, and Tian, 2025). Evidence also suggests that brand value targeted marketing by increasing consumer awareness, thereby incentivizing substantial firm investment in advertising (Cavenaile, Celik, Perla, and Roldan-Blanco, 2025; Baslandze, Greenwood, Marto, and Moreira, 2023). On the supply side, I conceptualize organizational capital as managerial productivity, including the firm's embodied managerial talent and its contribution to future production profitability (Carlin, Chowdhry, and Garmaise, 2012; Eisfeldt and Papanikolaou, 2013; Prescott and Visscher, 1980)

[1] The literature also categorizes software as a firm-specific intangible. For the sake of brevity, in this research I take the position that this type of intangible can diffuse from one firm to another through worker mobility. For knowledge diffusion through worker mobility and scientist networks, see Akerman and Holzheu (2025) and Zacchia (2020).

[2] Alternatively, Pearce and Wu (2025) suggest that brand value is transferable between firms. This transfer could occur through mergers and acquisitions (M&A). Although this may affect the transitional period in the economy, in the long run, brand value cannot accumulate infinitely through M&A because eventually all firms will exit.

Table 1. Taxonomy of Intangibles

	Supply Side	**Demand Side**
Embedded	Organizational Capital	Brand Value
Transferable	Software, Patents	

I build the theoretical framework by integrating two types of endogenous growth models: the step-by-step innovation model with vertical competition between two firms in the Akcigit and Ates (2023), and the Peters (2020) model[3], which allows firms to engage in both horizontal and vertical innovation. I combine these two frameworks to capture competition and creative destruction process along both the horizontal and vertical dimensions. While each of these models features only a single type of intangible investment, I extend them by allowing for two distinct types, capturing firms' choices between transferable and embedded intangible investments. Furthermore, I introduce span-of-control frictions that arise as firms expand horizontally across more industries.

The economy consists of a single final-good sector and a continuum of intermediate-good sectors. In each intermediate goods sector, a single superstar firm and a continuum of fringe firms produce intermediate goods for the final goods sector. Superstar firms can invest in embedded intangibles, which improve managerial productivity and the perceived quality of their products through brand value and organizational capital. Alternatively, they can invest in transferable intangibles, which enhance product quality in two ways: (i) by vertically improving existing products and (ii) by expanding their product portfolio through entry into new sectors. Importantly, vertical improvements do not create span-of-control constraints; however, horizontal expansion into new sectors splits managerial attention, therefore creating a span of control friction (Lucas, 1978)[4].

[3]These models provide modern interpretations of two canonical strands of the endogenous growth literature: the step-by-step innovation model developed by Aghion, Harris, Howitt, and Vickers (2001) and Aghion, Bloom, Blundell, Griffith, and Howitt (2005), and the horizontal innovation framework of Klette and Kortum (2004).

[4]See also Jovanovic (2025). Alternatively, Acemoglu, Akcigit, Alp, Bloom, and Kerr (2018) uses skilled labor in operational activities as a fixed cost, creating a trade-off between its allocation to R&D and expanding scope. While an increase in firm scale raises operational demands, this does not directly reduce firms' efficiency or pricing power with scope.

Also, these two types of intangibles differ in their scalability: while superstar firms can transfer their brand value and organizational capital freely across all sectors they operate in, quality improvements from transferable intangibles require separate investments in each new sector. This distinction implies that embedded investments exhibit increasing returns to scale as firms expand horizontally since the benefits of brand and organizational capital accumulate across sectors without proportional increases in cost.

Fringe firms produce homogeneous products and cannot accumulate brand value or organizational capital. When a superstar firm exits a sector, its transferable intangibles—the established product quality in that sector—become freely available to fringe firms. For these fringe firms, the only path to becoming a superstar is through radical innovation. Under oligopolistic Bertrand competition between superstar and fringe firms, a superstar's markup in each sector is endogenously determined by three factors: the product-quality gap relative to fringe firms, the superstar's stock of firm-specific intangibles, and the number of sectors in which it operates.

Creative destruction in the model can arise from two distinct sources: a superstar firm from another industry may challenge the incumbent in this sector, or a fringe firm may capture it through drastic innovation. However, a higher level of embedded intangibles lowers the incumbent superstar's marginal cost, making it more difficult for other firms to enter through innovation. As a superstar firm expands into more sectors, managerial attention must be distributed across a broader set of activities. This creates a classic span-of-control problem: each individual sector receives less focused oversight, so per-sector managerial productivity declines with firm scope. At the same time, operating in more sectors also means the firm has accumulated a larger stock of managerial talent, organizational routines, and transferable expertise. When a competitive threat arises, the incumbent can temporarily reallocate and concentrate part of this broader managerial bench on the contested sector. Consequently, although normal operations suffer from lower per-sector managerial quality due to the dispersion of attention, the firm's total defensive capacity can still increase with its scope if coordination costs are not too severe. Thus, embedded intangibles and multi-sector presence jointly create an entry barrier for

both challenger superstars and fringe firms.

The model predicts that horizontal expansion decreases managerial productivity and pricing power within each sector due to span-of-control constraints. To offset these inefficiencies, multi-sector firms leverage cross-product synergies and shift investment toward embedded intangibles, which deliver increasing returns to scale. This strategic shift extends the life cycle of superstar firms in existing markets but generates adverse aggregate consequences: (i) contraction of innovation possibilities for new entrants through creative destruction; and (ii) depressed long-run quality improvements due to lower investment in transferable intangibles.

To empirically validate the model's predictions, I use Compustat fundamentals dataset to measure transferable-to-embedded intangible ratio, markups, and productivity. After that, I merge with the Compustat Segment dataset to measure firm scope via the 2-digit industries in which firms operate. I document a robust fact: as firms' scope increases, their markups, productivity, growth rates, and the ratio of transferable-to-embedded intangible investment all decline. To quantify entry barrier (a proxy for competitive pressure), I use the product-market fluidity dataset from Hoberg, Phillips, and Prabhala (2014) and show that, on average, firms face lower competitive threats and operate in less fluid markets as scope rises. I also employ the patent dataset from Kogan, Papanikolaou, Seru, and Stoffman (2017), which measures stock market reactions to granted patents. I use these data for two purposes. First, I use patent value as an innovation shock in local projections to identify robust causal effects. Second, I use forward citations and value per patent as proxies for the social benefits and private value of innovation, respectively, to examine how firms' generation of private value and social benefit evolves with firm scope.

I discipline the key parameters of the model using the simulated method of moments. Based on the calibrated model, I examine how individual superstar firms choose between vertical and horizontal innovation based on their comparative advantage, which is determined by the quality gap—proportional to the transferable intangible gap between the superstar and fringe firms—and by the level of embedded intangibles. When the quality gap between the superstar and fringe firms is large, superstar firms have a strong incen-

tive to pursue vertical innovation. On the other hand, once a superstar firm's embedded intangible level increases, it begins to engage in horizontal innovation. Without embedded intangibles, entering a new sector is costly for superstar firms due to span-of-control constraints. Absent such intangibles to mitigate these costs, superstar firms have no incentive to pursue horizontal innovation.

Next, I ask how the relative shares of brand value (demand-side) and organizational capital (supply-side) shape firm dynamics. An increase in the share of brand value relative to organizational capital induces firms to concentrate production within a single line, as brand value alone cannot offset the managerial frictions associated with expansion. Consequently, markups decline: the reduction in organizational capital lowers managerial productivity and outweighs demand-side gains from brand value. This shift also raises entry barriers for potential entrants, thereby reducing market fluidity and aggregate growth. In contrast, when organizational capital constitutes a larger share, firms are more likely to expand, as organizational capital directly enhances managerial efficiency. This fosters higher markups, greater market fluidity, and faster economic growth.

In the following, I run two counterfactuals: I remove the span-of-control constraint and eliminate embedded intangibles to isolate their effects on markups, firm size, market fluidity, and aggregate growth. Without span-of-control constraints, a firms to operate more product lines, raise average markups, increase investment in embedded intangibles (thereby reducing market fluidity), and raise aggregate growth. Eliminating embedded intangibles concentrates production on a single line, lowers markups modestly (0.5–1.5%), and raises aggregate growth via a large increase in creative-destruction-driven fluidity.

In the last part of the quantitative analysis, I examine how misallocation operates through two channels: (i) markup dispersion and (ii) entry barriers created by embedded intangibles. Pure markup dispersion accounts for only 0.5% of output loss. In contrast, removing entry barriers more than triples aggregate output; this is driven primarily by substantial quality-improvement gains, during which the contribution of embedded capital falls slightly. Motivated by these results, I evaluate three tax experiments: a size-dependent profit tax (10% → 12.5%), a flat 11.3% tax on embedded and expansion investment, and joint

application of all three taxes. In consumption-equivalent welfare, the size tax delivers the largest gain (+10.915%); the embedded-investment taxes yield modest gains (+1.745% and +2.176%); and the joint policy produces the most significant complementary benefit (+16.237%).

Related Literature. This paper contributes to the growing literature on intangibles and their effects on firm dynamics, suggesting that intangibles increase market concentration and markups, while reducing investment in tangible capital (Chiavari and Goraya, 2025; Crouzet and Eberly, 2019; Weiss, 2020). De Ridder (2024) conceptualizes software intangibles as firm-specific fixed costs and shows how incumbents' strategic investments lower marginal production costs, creating asymmetric barriers to innovation that favor incumbents. Similarly, Aghion, Bergeaud, Boppart, Klenow, and Li (2023) distinguishes between product and process (firm-specific) innovation and highlights the roles of information and communication technologies in driving concentration. In both setups, intangibles are treated as an inborn firm ability rather than a developed feature. Related to intangible heterogeneity, Cavenaile and Roldan-Blanco (2021) and Cavenaile, Celik, Roldan-Blanco, and Tian (2025) show that advertising can substitute for R&D and dampen innovation intensity, while Pearce and Wu (2025) examines brand-value transfer between firms in the context of market concentration. I contribute to the literature by proposing a unified, generalizable taxonomy of intangibles and by modeling their accumulation as an endogenous outcome of firm optimization, rather than treating them solely as expenses or inborn ability. Therefore, I provide microfoundations for firms' investment and accumulation decisions across different types of intangibles, delivering novel insights into how these choices shape intangible heterogeneity, innovation, and competition.

Firms' innovation direction is often explained in the literature through differences in firm size. Akcigit and Kerr (2018) develop an endogenous growth model in which incumbents choose between internal and external innovation with heterogeneous returns, and they show that smaller firms are more likely to engage in external innovation, whereas larger firms tend to focus on vertical, quality-improving innovation. Using French administrative data, Berlingieri, De Ridder, Lashkari, and Rigo (2025) document that firms

frequently expand through sequential product diversification rather than by first improving their existing products. In contrast, Garcia-Macia, Hsieh, and Klenow (2019) find that most innovation within firms arises from incumbents improving existing products instead of entering entirely new product lines. This paper contributes to the literature by moving beyond size-based explanations and proposing intangible heterogeneity as a central determinant of innovation direction. I show that firms of similar size may optimally choose different innovation paths depending on the composition of their intangible endowments. Firm size remains an important factor shaping innovation incentives, but it does not fully account for the observed heterogeneity in firms' innovation strategies. I therefore propose that intangible heterogeneity serves as an alternative mechanism driving innovation direction, alongside firm size.

This paper offers a complementary explanation for several documented trends: reduced knowledge spillovers (Akcigit and Ates, 2021; Akcigit and Ates, 2023), declining patent quality (Olmstead-Rumsey, 2019), production lock-in (Casal, 2024), and increasing entry barriers (Gutiérrez and Philippon, 2019). The mechanism centers on the shift in investment from transferable to embedded intangibles as firm scope expands. This change depresses spillovers due to the firm-specific nature of embedded intangibles. Furthermore, firms exercise broader control over their competitive environment, limiting the innovation potential of rivals and the broader diffusion of knowledge through their expanding scope and accumulation of embedded intangibles.

This paper contributes to the literature on resource misallocation (Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008) by identifying two distinct channels. First, misallocation can arise from markup dispersion (Peters, 2020; Edmond, Midrigan, and Xu, 2023), driven by firms' accumulation of both transferable and embedded intangibles. Second, the interaction between embedded intangibles and expanding firm scope creates entry barriers that amplify distortions and reduce allocative efficiency. I show that the joint increase in embedded intangibles and firm scope makes the process of creative destruction asymmetric, leading to persistent resource misallocation.

The empirical and theoretical literature on span-of-control constraints has focused

primarily on hierarchical organization, knowledge-flow frictions, managerial ability, and the associated premiums (Smeets, Waldman, and Warzynski, 2019; Bandiera, Prat, Sadun, and Wulf, 2014; Garicano, 2000; Bloom and Van Reenen, 2007). From a macro perspective, span-of-control constraints are closely linked to the firm-size distribution (Jovanovic, 2025). I disaggregate firm size into vertical and horizontal dimensions of innovation and show that only horizontal expansion generates meaningful span-of-control frictions. This approach allows me to examine how span-of-control constraint shapes firm dynamics and growth, highlighting its implications for innovation and investment decisions.

Outline. The remainder of the paper is organized as follows: Section 3 presents the datasets and empirical facts; Section 2 introduces the theoretical model and characterizes its equilibrium; Section 4 discusses the calibration; Section 5 examines counterfactual analysis; Section 6 analyzes misallocation and policy implications and Section 7 concludes.

2. Theoretical Model

By unifying the vertical innovation framework with the horizontal expansion mechanism, the model introduces two key elements: heterogeneity in intangible investment and a span-of-control constraint. These features jointly determine firms' investment decisions, markups, and output based on their competitive positions and the composition of their intangibles.

2.1. Economic Environment

Preferences. In this economy, continuous time is represented by t, and household preferences are described by a logarithmic utility function:

$$\int_0^\infty e^{-\rho t} \ln(C_t)\, dt, \tag{1}$$

where C_t represents household consumption, and $\rho > 0$ denotes the time discount rate. The budget constraint is expressed as

$$\dot{A}_t = r_t\, A_t + w_t - C_t. \tag{2}$$

The term A_t represents the total assets in the economy at time t, and the labor supply is normalized to 1. I normalize the price of the consumption good; therefore, w_t and r_t show the relative prices of wage and the interest rate, respectively. Because households own firms, total assets in the economy can be expressed as the sum of the firm values

$$A_t = \int_0^1 \left(V_{sjt} + V_{fjt}\right) dj,$$

where V_{sjt} and V_{fjt} denote the values of superstar and fringe firms in the intermediate good sector j at time t.

Final Good Technology and Market Structure. The final good sector used for consumption is produced according to the following technology:

$$\ln(Y_t) = \int_0^1 \ln(y_{jt})\, dj. \tag{3}$$

It is produced using a continuum of intermediate varieties $j \in [0, 1]$ in a perfectly competitive market. In each intermediate goods sector, one superstar firm, y_{sjt}, and a continuum of homogeneous fringe firms, y_{fjt}, compete à la Bertrand to supply the final good producer. Their output is aggregated by a constant elasticity of substitution:

$$y_{jt} = \left(\underbrace{\frac{\chi(e_{st})}{1+\chi(e_{st})}}_{\substack{\Xi(e_{st}) \\ \text{Perceived Quality}}} \; y_{sjt}^{\varepsilon} + \underbrace{\frac{1}{1+\chi(e_{st})}}_{\frac{\Xi(e_{st})}{\chi(e_{st})}} \; y_{fjt}^{\varepsilon} \right)^{\frac{1}{\varepsilon}}, \tag{4}$$

where $\varepsilon \in (0, 1)$. In each production line j, a superstar firm s may own multiple sectors. It is characterized by the countable set of sectors for which it owns the leading technology,

$J_s \subseteq [0,1]$. The number of leading sectors owned by superstar firms s is given by $n_s = |J_s| \in \mathbb{Z}_+$.

In each sector, superstar firms produce a differentiated product; the term $\chi(\xi e_{st})$ is an endogenous and concave demand shifter defined as $\chi(\xi e_{st}) = (\xi e_{st})^{\beta}$. Here, e_{st} represents the embedded intangibles of firm s while ξe_{st} denotes the portion of embedded intangibles associated with brand value, with $\xi \in (0,1)$. The parameter $\beta \in (0,1)$ captures the curvature of the demand shifter. For simplicity, the embedded intangible level and brand value of fringe firms are normalized to one. The term $\Xi(e_{st})$ captures the notion that an increase in a superstar firm's relative brand value raises the perceived quality (benefit) of its product in the final-good sector relative to fringe firms. Finally, fringe firms are homogeneous within each intermediate sector and can be represented as:

$$y_{fjt} = \int_0^1 y_{ijt}\, di, \tag{5}$$

where each fringe firm $i \in (0,1)$.

Superstar Firm Production. The production function for superstar firm s in line j at time t is given by

$$y_{sjt} \;=\; \underbrace{q_{sjt}}_{\text{Product Quality}} \; \underbrace{\psi(e_{st}, n_{st})}_{\text{Managerial Quality}} \; \underbrace{l_{sjt}}_{\text{Labor Input}}\;, \tag{6}$$

$$\text{with} \quad \psi(e_{st}, n_{st}) \;=\; \frac{\big((1-\xi)\, e_{st}\big)^{\alpha}}{\gamma\, n_{st}^{\alpha}}.$$

The term $q_{sjt}\,\psi(\cdot)$ represents the total productivity of firm s in sector j, where q_{sjt} denotes product quality and $\psi(\cdot)$ captures the firm's managerial productivity. The input l_{sjt} is the quantity of labor employed by superstar firm s in sector j. The component $(1-\xi)e_{st}$ represents the fraction $(1-\xi)$ of embedded intangibles interpreted as organizational capital, which improves managerial efficiency. The variable n_s denotes the number of sectors owned by firm s. As n_s increases, managerial productivity per sector declines due to the span of control—expansion reduces the firm's ability to effectively oversee each sector[5].

[5]The span of control constraint imposes a natural upper bound $\bar{n}$ on horizontal firm expansion.

Furthermore, the curvature of organizational capital and the span of control constraint are governed by α, and the value of γ determines the scale parameter.

The summation of managerial productivity for superstar s across all active sectors[6] j is represented by

$$\Lambda \cdot e_{st}^{\alpha} \cdot n_{st}^{1-\alpha} \quad \text{with} \quad \Lambda \equiv \frac{(1-\xi)^{\alpha}}{\gamma}.$$

When $0 < \alpha < 1$, total managerial quality follows a Cobb-Douglas form: increasing both organizational capital and scope yields diminishing marginal returns to each factor. Conversely, when $\alpha > 1$, diseconomies of scope emerge: the coordination costs of managing multiple sectors outweigh the benefits of scale, and total managerial productivity decreases with firm scope. Total managerial productivity determines the firm's short-term defensive capacity against competitive threats, further details are provided in the section on creative destruction.

Fringe Firm Production. Fringe firms produce output using linear technology:

$$y_{fjt} = q_{fjt}\, l_{fjt}, \tag{7}$$

where q_{fjt} denotes productivity and l_{fjt} is labor input. Unlike superstar firms, a fringe firm operates in only one sector and has a managerial quality normalized to 1. Its productivity is inherited: when a superstar firm exits, its transferable intangible assets, such as patents, become publicly available, allowing a fringe firm to adopt the previous leader's productivity level.

Investment Functions and Innovation. A key assumption of the model is that quality improvements in each sector require separate investments, whereas embedded intangibles are freely mobile across sectors. Thus, a successful investment in embedded intangibles simultaneously improves the brand value and the organizational capital of all sectors within the firm. Under this framework, superstar firms face three investment decisions: they can enter a new sector or improve the quality of existing product through

[6]Managerial productivity is symmetric across sectors. Therefore, total managerial productivity equals the managerial productivity of the single sector multiplied by the number of active sectors of firm s

transferable intangibles, or improve brand value and organizational capital of all lines via embedded intangible investments. These investment scenarios are illustrated in Figure 1.

The variables $I_{j,t}^{\text{Emb}}$, $I_{j,t}^{\text{Ver}}$, and $I_{j,t}^{\text{Hor}}$ represent the investment of superstar s in embedded intangibles, vertical improvement investment on its own sector, and horizontal investment on new sector, respectively. Each unit of investment generates a successful flow rate of innovation on vertical $z_{j,t}^{\text{Ver}}$, horizontal $z_{j,t}^{\text{Hor}}$, or the embedded intangible level $z_{j,t}^{\text{Emb}}$, respectively. Investments occur with convex costs and are represented by

$$I_{jt}^{\text{Ver}} = \gamma^{\text{Ver}} \left(z_{sjt}^{\text{Ver}}\right)^{\vartheta^{\text{Ver}}} Y_t \quad , \quad I_{jt}^{\text{Hor}} = \gamma^{\text{Hor}} \left(z_{jt}^{\text{Hor}}\right)^{\vartheta^{\text{Hor}}} Y_t \tag{8}$$

$$\text{and} \quad I_{jt}^{\text{Emb}} = \gamma^{\text{Emb}} \left(z_{jt}^{\text{Emb}}\right)^{\vartheta^{\text{Emb}}} Y_t.$$

In the above expressions, the investment cost function scales with the size of the economy, Y_t. The parameters γ^{Ver}, γ^{Hor}, and γ^{Emb} determine the cost scale of the investment functions, whereas ϑ^{Ver}, ϑ^{Emb}, and ϑ^{Hor} determine their curvature, respectively. The total investment in transferable intangibles by a superstar firm in production line j at time t is equal to

$$I_{jt}^{\text{T}} = I_{jt}^{\text{Ver}} + I_{jt}^{\text{Hor}}. \tag{9}$$

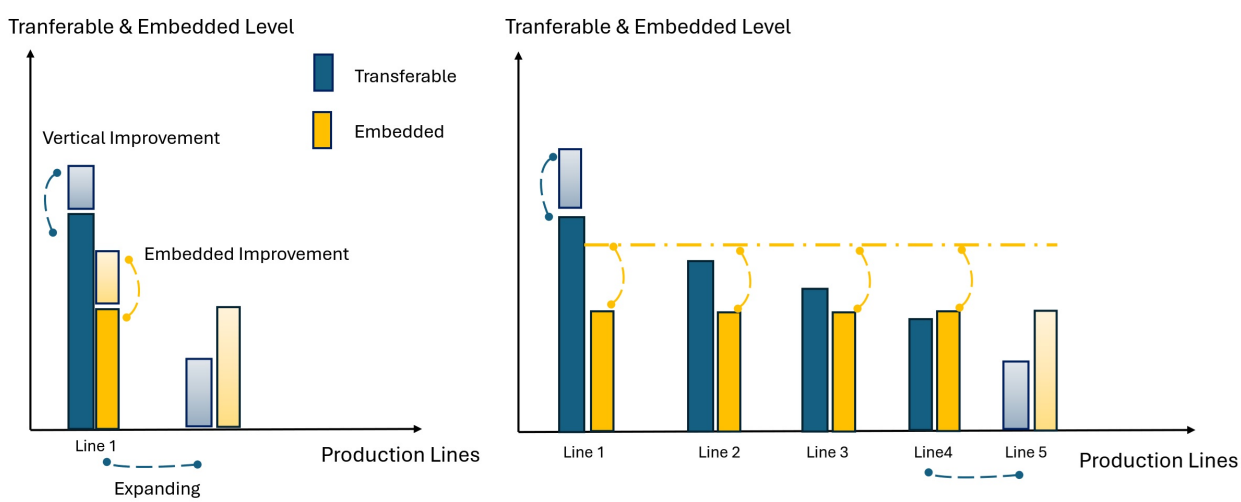

Figure 1. Firm Investment and Innovation Types

Fringe firms invest only in transferable intangibles within their production line, aim-

ing to achieve drastic innovations that could elevate them to become a new superstar. Fringe firms invest in transferable intangibles, followed by

$$I_f = \gamma^f \left(z_f\right)^{\vartheta^f} Y_t. \tag{10}$$

The term γ^f represents the fringe firms' cost of scale, and ϑ^f is the curvature of their investment. On each production line, firms invest in transferable and embedded intangibles to improve product quality or to increase brand value and managerial productivity. The dynamics are given by

$$q_{sjt} = \lambda^{m_{sjt}} q_{sj0}, \quad \text{and} \quad e_{st} = \theta^{k_{st}} e_{j0}, \tag{11}$$

with initial levels $q_{sj0} = 1$ and $e_{j0} = 1$. The variables m_{sjt} and k_{st} denote the cumulative numbers of product-quality and managerial/brand-improving innovations by firm s on line j up to time t, respectively. When a firm successfully innovates between t and $t + \Delta t$, its quality increases by a factor of $\lambda > 1$. Brand value and managerial productivity evolve similarly; however, improvement by a factor of $\theta > 1$ is firm-specific and depends only on firm s.

Upon the exit of a superstar firm, its transferable technology becomes imitable, allowing fringe firms on production line j to adopt it. Consequently, the quality gap between a superstar firm and a fringe firm in line j at time t can be expressed as

$$\frac{q_{sjt}}{q_{fjt}} = \frac{\lambda^{m_{sjt}}}{\lambda^{m_{fjt}}} = \lambda^{m_{sjt} - m_{fjt}} = \lambda^{m_{jt}}, \tag{12}$$

where $m_{jt} \equiv m_{sjt} - m_{fjt}$ denotes the technology gap in transferable intangibles. Because the embedded level of fringe firms on each production line is normalized to one, the gap in brand value and managerial quality is given by the superstar's embedded stock:

$$\frac{e_{sjt}}{e_{fjt}} = \frac{\theta^{k_{sjt}}}{1} = \theta^{k_{st}}, \tag{13}$$

where k_{st} represents the gap in embedded intangibles. To ensure a finite state space, I impose upper bounds $\bar{m}$ and $\bar{k}$ on the transferable gap m_{jt} and the embedded gap k_{st}, respectively.

Creative Destruction. Both superstar and fringe firms engage in creative destruction via product quality innovations. Their takeover dynamics, however, differ significantly due to asymmetric capabilities in innovation and types of intangible accumulation.

Assumption (Managerial Reallocation Capability)**.** *A superstar firm can temporarily reallocate its total managerial capacity from its entire scope to a single sector when engaging in Bertrand price competition with a challenger firm. This reallocation is costless in the short term but affects the price competition stage in that sector.*

Challenger Superstar Creative Destruction. When a challenger superstar makes an incremental innovation that improves product quality by a factor λ, it can become the new superstar if its marginal cost falls below that of the incumbent:

$$\frac{w_t}{\Lambda\, q_{cjt}\, e_{ct}^{\alpha}\, n_{ct}^{1-\alpha}} < \frac{w_t}{\Lambda\, q_{sjt}\, e_{st}^{\alpha}\, n_{st}^{1-\alpha}}.$$

Since the innovation improves quality such that $q_{cjt} = \lambda\, q_{sjt}$, and the challenger randomly enters sector j', the probability of successful entry is

$$\sum_{m}\sum_{k}\sum_{n_s} \mathbb{I}\Big\{\lambda\,(\theta^{k'})^{\alpha}\, n_{s't}^{1-\alpha} > (\theta^{k})^{\alpha}\, n_{st}^{1-\alpha}\Big\}\mu(m,k,n) \equiv \mathbb{P}_{s'>s}\,,$$

where the left-hand side represents the challenger's innovation and total managerial productivity and the right-hand side represents the incumbent's total managerial productivity. This comparison determines whether the challenger can produce at a sufficiently low cost to capture the market in the short term. The term μ shows mas of firm in state (m,k,n).

Fringe Firms Creative Destruction. Fringe firms cannot accumulate embedded intangibles; therefore, their only path to becoming a new superstar is through radical innova-

tion[7]. This entails jumping multiple steps on the quality ladder—specifically, improving quality by a factor $\bar{\lambda}$ rather than a single step, such that $q^f_{cjt} = \bar{\lambda}\, q_{sjt}$. A fringe firm successfully becomes the new superstar if its marginal cost falls below that of the incumbent, which requires

$$\mathbb{I}\left\{\bar{\lambda} > e^{\alpha}_{st}\, n^{1-\alpha}_{st}\right\} \equiv \mathbb{I}_{f>s}.$$

2.2. Equilibrium

This section characterizes the general equilibrium of the model, which consists of a static and a dynamic component. The analysis begins with the static equilibrium, determining prices and allocations for a given set of states. Subsequently, I define the Markov Perfect Equilibrium for the dynamic game, outlining the value functions, optimal policy functions, and the evolution of the aggregate state distribution.

Household's Problem. A household maximizes utility with the Euler Equation:

$$\frac{\dot{C}_t}{C_t} = r_t - \rho. \tag{14}$$

Along the balanced growth path, consumption and output grow at the same rate, $g = r - \rho$, and the transversality condition holds.

Final and Intermediate Good Sectors. The final-good producer's demand for the continuum of intermediate goods on line j satisfies

$$p_{jt} \;=\; \frac{Y_t}{y_{jt}}. \tag{15}$$

This implies the demand functions for the superstar and the fringe firms:[8]

$$y_{sjt} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t \big(\chi(e_s)\big)^{\frac{1}{1-\varepsilon}} \quad \text{and} \quad y_{fjt} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{fjt}^{\frac{1}{\varepsilon-1}} Y_t, \tag{16}$$

[7] Fringe firms in the model correspond to small, young firms in the data. A vast literature documents that small and young firms grow faster than their larger counterparts, see Evans (1987); Haltiwanger, Jarmin, and Miranda (2013).

[8] See Appendix B.2 for the full derivations

where p_{sjt} and p_{fjt} represent the product prices of the superstar and fringe firms, respectively. Furthermore, p_{jt} is the ideal price index for production line j, given by the following equation:

$$p_{jt} = \left(\left(\chi(e_s)\right)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}}\right)^{\frac{\varepsilon-1}{\varepsilon}}. \tag{17}$$

Prices and Market Share Function. The Cobb-Douglas production function for the final good indicates equal expenditure shares across all production lines. The market share of superstar firm s in industry j at time t is defined as

$$\frac{p_{sjt}y_{sjt}}{p_{jt}y_{jt}} = \frac{p_{sjt}y_{sjt}}{Y_t} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} \chi(e_s)^{\frac{1}{1-\varepsilon}} \equiv \phi_{sjt}. \tag{18}$$

Because the sum of market shares equals one, the fringe firms' share is $1 - \phi_{sjt}$. The equilibrium price of the superstar firm on production line j at time t under competition à la Bertrand [9] is then given by:[10]

$$p_{sjt} = \frac{1 - \varepsilon\,\phi_{sjt}}{(1 - \phi_{sjt})\,\varepsilon}\, MC_{sjt}, \tag{19}$$

where $MC_{sjt} = w_t \times \dfrac{\gamma_s\, n_{st}^{\alpha_s}}{q_{sjt}\left((1-\xi)e_{st}\right)^{\alpha}}$ denotes the marginal cost of the superstar firm, and the fringe firms' price is equal to their marginal cost[11]. The superstar firm's price equation demonstrates that its price is positively correlated with product quality and embedded intangible level, while it is inversely related to the number of production lines operated. The price ratio of fringe firms to superstar follows

$$\frac{p_{fjt}}{p_{sjt}} = \frac{(1-\phi_{sjt})\varepsilon}{1-\varepsilon\phi_{ijt}} \cdot \lambda^{m_j} \left(\frac{(1-\xi)e_{st}}{n_s}\right)^{\alpha}. \tag{20}$$

Using the definition of market share in equation (18) and substituting the ideal price index

[9] For the Cournot Competition version, see Appendix B.4

[10] See Appendix B.3 for the full derivations

[11] Even though fringe firms lack independent pricing power, their presence creates a competitive constraint that disciplines superstar firms. This competitive pressure forces superstars to engage in limit pricing strategies, preventing them from fully exercising their market power and extracting monopolistic rents.

from equation (17), the market share of the superstar firm can be expressed in terms of relative prices as

$$\phi_{sjt} = \frac{1}{1+\left(\frac{1}{(\xi e_{st})^{\frac{\beta}{1-\varepsilon}}}\left(\frac{p_{fjt}}{p_{ijt}}\right)^{\frac{\varepsilon}{\varepsilon-1}}\right)}. \tag{21}$$

Replacing the relative price ratio with equation (20) shows that the superstar firm's market share depends on the quality gap m_j, embedded intangible level e_{st}, and the number of production lines n_s it operates.

Profit, Markup and Labor Demand. The static operational profit of the superstar firm is proportional to its market share and the size of the economy:

$$\pi_{sjt} = \frac{(1-\varepsilon)\,\phi_{sjt}}{1-\varepsilon\phi_{sjt}}\,Y_t, \tag{22}$$

with the corresponding markup given by

$$\sigma_{sjt} = \frac{1-\varepsilon\phi_{sjt}}{(1-\phi_{sjt})\,\varepsilon}. \tag{23}$$

Both markup and profit increase with market share. However, because market share on a given line is inversely related to the number of lines a firm operates, a superstar's markup and profit decrease, reflecting the natural consequence of diminishing managerial productivity across production lines. The optimal labor inputs for superstar and fringe firms, respectively, are[12]

$$l_{sjt} = \frac{\phi_{sjt}}{\sigma_{sjt}}\,\omega_t^{-1}, \tag{24}$$

$$l_{fjt} = (1-\phi_{sjt})\,\omega_t^{-1}, \tag{25}$$

where $\omega_t = \frac{w_t}{Y_t}$ denotes the wage share of the economy.

The static equilibrium provides only an implicit solution. Nevertheless, the model yields tractable dynamics because the equilibrium outcome for a superstar firm depends

[12]See equation (65) in the Appendix for details.

solely on its market share, which is in turn determined by the quality gap, the level of embedded intangibles, and the number of production lines operated. This tractability makes it possible to analyze firms' endogenous investment decisions in different types of intangibles and to study how these choices affect firm dynamics.

Superstar Value Function. The superstar value function $V_t(\mathbf{m}, k, n)$ relevant payoff depends on the quality gap vector $\mathbf{m} = \{m_j\}_{j=1}^n$, embedded intangible gap k, and the number of sectors the superstar firm has n[13]. Superstar firm s optimizes the flow rate of innovation of z_{jt}^{Ver}, z_{jt}^{Hor}, and z_{jt}^{Emb} to maximize the value function given by equation 26.

$$
\begin{aligned}
r_t V_t(\mathbf{m}, k, n) - \dot{V}_t(\mathbf{m}, k, n) = \max_{z_{jt}^{\text{Ver}}, z_{jt}^{\text{Hor}}, z_{jt}^{\text{Emb}}} \sum_{j=1}^{n} \Bigg(& \pi_{jt}(m_j, k, n) \\
& + \underbrace{\mathbb{P}_{k \geq k'} z_{jt}^{Hor} \Big[V_t\big((m_j, 1), k, n+1\big) - V_t(m_j, k, n) \Big]}_{\text{Horizontal Innovation}} + \underbrace{z_{jt}^{\text{Ver}} \Big[V_t(m_j + 1, k, n) - V_t(m_j, k, n) \Big]}_{\text{Vertical innovation}} \\
& + \underbrace{z_{jt}^{\text{Emb}} \Big[V_t(\mathbf{m}, k+1, n) - V_t(\mathbf{m}, k, n) \Big]}_{\text{Innovation on Embedded}} + \underbrace{\mathbb{P}_{k' \geq k} Z_{jt}^{\text{Hor}} \Big[V_t(m_j, k, n-1) - V_t(m_j, k, n) \Big]}_{\text{Challenger Superstar innovation}} \\
& + \underbrace{\mathbb{I}_{f > s} Z_{jt}^{f} \Big[V_t(m_j, k, n-1) - V_t(m_j, k, n) \Big]}_{\text{Fringe Firms' Innovation}} - \gamma^{\text{Ver}} \left(z_{jt}^{\text{Ver}} \right)^{\vartheta^{\text{Ver}}} Y_t - \gamma^{\text{Emb}} \left(z_{jt}^{\text{Emb}} \right)^{\vartheta^{\text{Emb}}} Y_t \\
& - \gamma^{\text{Hor}} \left(z_{jt}^{\text{Hor}} \right)^{\vartheta^{\text{Hor}}} Y_t \Bigg)
\end{aligned}
\tag{26}
$$

The left-hand side of the value function shows the return on the value function and its gain over time. The first line on the right-hand side represents the profit of the superstar firm in sector j. The second line's first term shows that, with an innovation flow rate of $\mathbb{P}_{k \geq k'} z_{jt}^{Hor}$, firm s increases its scope from n_s to $n_s + 1$, while the subsequent terms describe to increase sector j's quality gap by one rung with an innovation flow rate of z_{jt}^{Ver}. The third line shows that superstar firm s can increase their embedded intangible level by one rung across all production lines, and the next terms show that challenger superstars and fringe firms can innovate, and superstar s exits sector j. The last three terms

[13]To simplify notation, the s subscript denoting the superstar firm is omitted whenever it is clear from context.

represent the cost of investment in improving existing production quality, improving the embedded intangible level, and entering a new sector, respectively. Cost function details are described in equations (8).

The terms Z_{jt}^{Hor} and Z_{jt}^{f} denote aggregate horizontal innovation by challenger superstar firms and aggregate innovation by fringe firms:

$$Z_{jt}^{\text{Hor}} = \int_0^1 z_{sjt}^{\text{Hor}}\, dj, \qquad Z_{jt}^{f} = \int_0^1 z_{ijt}\, di.$$

In the balanced growth path, aggregate output Y, consumption C, and the value function $V(\mathbf{m}, k, n)$ all grow at the constant rate g. Defining the stationary value function as $v(\mathbf{m}, k, n) = \frac{V(\mathbf{m},k,n)}{Y}$, superstar firm value function on the balanced growth path is given by:

$$\begin{aligned}
\rho v(\mathbf{m}, k, n) = \max_{z_j^{\text{Ver}}, z_j^{\text{Hor}}, z_j^{\text{Emb}}} \sum_{j=1}^{n} \Bigg(& \pi_j(m_j, k, n) \\
& + \underbrace{\mathbb{P}_{k \geq k'}\, z_j^{Hor} \Big[v\big((m_j, 1), k, n+1\big) - v(m_j, k, n) \Big]}_{\text{Horizontal Innovation}} + \underbrace{z_j^{\text{Ver}} \Big[v(m_j + 1, k, n) - v(m_j, k, n) \Big]}_{\text{Vertical innovation}} \\
& + \underbrace{z_j^{\text{Emb}} \Big[v(\mathbf{m}, k+1, n) - v(\mathbf{m}, k, n) \Big]}_{\text{Innovation on Embedded}} + \underbrace{\mathbb{P}_{k' \geq k}\, Z_j^{\text{Hor}} \Big[v(m_j, k, n-1) - v(m_j, k, n) \Big]}_{\text{Challenger Superstar innovation}} \\
& + \underbrace{\mathbb{I}_{f > s}\, Z_j^{f} \Big[v(m_j, k, n-1) - v(m_j, k, n) \Big]}_{\text{Fringe Firms' Innovation}} - \gamma^{\text{Ver}} \left(z_j^{\text{Ver}} \right)^{\vartheta^{\text{Ver}}} - \gamma^{\text{Emb}} \left(z_j^{\text{Emb}} \right)^{\vartheta^{\text{Emb}}} \\
& - \gamma^{\text{Hor}} \left(z_j^{\text{Hor}} \right)^{\vartheta^{\text{Hor}}} \Bigg)
\end{aligned} \tag{27}$$

Innovation Decisions. The first-order conditions of the superstar value function determine optimal innovation intensities. Along the balanced growth path, the superstar firm's

optimal rates of vertical, embedded, and horizontal innovations are given by

$$z_j^{\text{Ver}} = \left(\frac{\Big[v(m_j+1,k,n) - v_t(m_j,k,n)) \Big]}{\gamma^{\text{Ver}} \cdot \vartheta^{\text{Ver}}} \right)^{\frac{1}{\vartheta^{\text{Ver}}-1}}, \tag{28}$$

$$z_j^{\text{Hor}} = \left(\frac{\mathbb{P}_{k \geq k'} \Big[v\big((m_j,1),k,n+1\big) - v(m_j,k,n) \Big]}{\gamma^{\text{Hor}} \cdot \vartheta^{\text{Hor}}} \right)^{\frac{1}{\vartheta^{\text{Hor}}-1}}, \tag{29}$$

$$z_j^{\text{Emb}} = \left(\frac{\Big[v(\mathbf{m},k+1,n) - v(\mathbf{m},k,n) \Big]}{\gamma^{\text{Emb}} \cdot \vartheta^{\text{Emb}}} \right)^{\frac{1}{\vartheta^{\text{Emb}}-1}}. \tag{30}$$

The value function of a fringe firm depends on its innovation intensity z_{ft} and on the indicator function $\mathbb{I}_{f>s}$, which captures whether drastic innovation is sufficiently large to displace the incumbent superstar.

$$r_t V_{ft}(m,k,n) - \dot{V}_{ft}(m,k,n) = \max_{z_{ft}} \left\{ \mathbb{I}_{f>s}\, z_{ft} \Big[V_f(\bar{\lambda},1,1) - V_f(m,k,n) \Big] - \gamma^f (z_{ft})^{\vartheta^f} Y_t \right\} \tag{31}$$

along the balanced growth path, the value function satisfies

$$\rho\, v_f(m,k,n) = \max_{z_f} \left\{ \mathbb{I}_{f>s}\, z_f \Big[v_f(\bar{\lambda},1,1) - v_f(m,k,n) \Big] - \gamma^f (z_f)^{\vartheta^f} \right\} \tag{32}$$

the optimal innovation intensity is therefore

$$z_f = \left(\frac{\mathbb{I}_{f>s} \left[v_f(\bar{\lambda},1,1) - v_f(m,k,n) \right]}{\gamma^f \vartheta^f} \right)^{\frac{1}{\vartheta^f-1}}. \tag{33}$$

Distribution Evolution. For notational simplicity, I suppress explicit indices for firm s,

industry j.

$$
\begin{aligned}
\mu_t(m,k,n) = {} & z_t^{\text{Ver}}(m-1,k,n)\,\mu_t(m-1,k,n) + z_t^{\text{Emb}}(m,k-1,n)\,\mu_t(m,k-1,n) \\
& + \mathbb{P}_{k\geq k'}\, z_t^{\text{Hor}}(m,k,n-1)\,\mu_t(m,k,n-1) \\
& + z_t^{\text{Ver}}(m-1,k-1,n)\, z_t^{\text{Emb}}(m-1,k-1,n)\,\mu_t(m-1,k-1,n) \\
& + z_t^{\text{Ver}}(m-1,k,n-1)\,\mathbb{P}_{k\geq k'}\, z_t^{\text{Hor}}(m-1,k,n-1)\,\mu_t(m-1,k,n-1) \\
& + z_t^{\text{Emb}}(m,k-1,n-1)\,\mathbb{P}_{k\geq k'}\, z_t^{\text{Hor}}(m,k-1,n-1)\,\mu_t(m,k-1,n-1) \\
& + z_t^{\text{Ver}}(m-1,k-1,n-1)\, z_t^{\text{Emb}}(m-1,k-1,n-1) \\
& \times\ \mathbb{P}_{k\geq k'}\, z_t^{\text{Hor}}(m-1,k-1,n-1)\,\mu_t(m-1,k-1,n-1) \\
& - z_t^{\text{Ver}}(m,k,n)\,\mu_t(m,k,n) - z_t^{\text{Emb}}(m,k,n)\,\mu_t(m,k,n) \\
& - \mathbb{P}_{k\geq k'}\, z_t^{\text{Hor}}(m,k,n)\,\mu_t(m,k,n) - \mathbb{I}_{f>s}\, Z_t^{f}(m,k,n)\,\mu_t(m,k,n) - \mathbb{P}_{k'\geq k}\, Z_t^{\text{Hor}}\,\mu_t(m,k,n)
\end{aligned}
\tag{34}
$$

The law of motion of the distribution is driven by net flows into and out of the cohort in state (m,k,n): inflows consist of firms that, following successful innovation, enter (m,k,n); outflows consist of firms that leave (m,k,n) as a result of their own innovation or innovation elsewhere. Concretely, the first three terms on the right-hand side capture inflows from predecessor states that are one step behind in a single dimension. The next three terms capture inflows from states that are one step behind in two dimensions (pairwise lags), while the subsequent term captures inflows from firms that are simultaneously one step behind in all three dimensions. The following group of terms describes outflows from state (m,k,n) owing to the superstar firm's own vertical, embedded, or horizontal innovation. Finally, the last two terms account for outflows induced by innovation by fringe firms or by leaders in other industries.

Distribution at Boundaries. The state space is bounded above and below, so specific conditions apply at its edges. If any state attains an upper bound, $m=\bar{m}$, $k=\bar{k}$, or $n=\bar{n}$ (including any joint combination), the outflow resulting from a successful innovation that would advance the firm further in that dimension is zero. This reflects the assumption that a firm's lead in any dimension cannot exceed a technologically feasible maximum. Conversely, if any state attains a lower bound, $m=1$, $k=1$, or $n=1$ (or any joint combination), the outflow to a state with a lower value in that dimension is zero. This condition

functions as an absorbing barrier, preventing a firm's position from deteriorating below a fundamental minimum.

Aggregate Variables. The joint distribution of (m, k, n) satisfies

$$\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \mu_t(m, k, n) = 1. \tag{35}$$

The labor market clears:

$$1 = \int_0^1 \left(l_{sjt} + l_{fjt}\right) dj, \tag{36}$$

and, using (24), (25) and (36), the normalized wage is

$$\omega_t = \sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(\tfrac{\phi_t(m,k,n)}{\sigma_t(m,k,n)} + 1 - \phi_t(m, k, n)\right) \mu_t(m, k, n). \tag{37}$$

Combining intermediate good sectors output (3), (6), (7) with the labor demand of superstar (24) and fringe firms (25) produces aggregate output

$$Y_t = Q_t\, \omega_t^{-1} \exp\left(\textstyle\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \underbrace{\ln \left[\left(\xi\theta^{k_t}\right)^{\beta} \left(\frac{((1-\xi)\theta^{k_t})^{\alpha}}{\gamma n_t^{\alpha_s}} \frac{\phi_t(m,k,n)}{\sigma_t(m,k,n)} \right)^{\varepsilon} + \left(\lambda^{-m_t}(1-\phi_t(m,k,n))\right)^{\varepsilon} \right]^{\frac{1}{\varepsilon}}}_{\equiv R_t(m,k,n)} \mu_t(m,k,n) \right). \tag{38}$$

where

$$Q_t \;=\; \exp\left(\int_0^1 \ln q_{sjt}\, dj\right)$$

The growth rate of the economy is[14]

$$g_t = -g_{\omega,t} + g_{Q,t} + g_{R,t}. \tag{39}$$

In a state of balanced growth, the economy grows at rate g, which is given by

$$g \;=\; \ln \lambda \sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(z^{\text{Int}}(m, k, n) + \mathbb{P}_{k \geq k'}\, z^{\text{Hor}}(m, k, n) + Z^f(m, k, n) \right) \mu(m, k, n). \tag{40}$$

[14]For details of the growth-rate calculation, see Appendix B.5.

Finally, the resource constraint satisfies

$$Y_t = C_t + \int_0^1 \left(I_{jt}^{\text{Ver}} + I_{jt}^{\text{Hor}} + I_{jt}^{\text{Emb}}\right) dj + \int_0^1 I_{jt}^f \, dj, \tag{41}$$

with

$$I_{jt}^f = \int_{F_j} I_{ijt} \, di.$$

Equilibrium Definition. The Markov Perfect equilibrium of the economy consists of an allocation $\{C_t, Y_t, y_{sjt}, y_{fjt}\}$, prices $\{r_t, w_t, p_{sjt}, p_{fjt}\}$, and policies $\{z^{\text{Ver}}, z^{\text{Hor}}, z^{\text{Emb}}, z_f, l_{sjt}, l_{fjt}\}$, such that the final goods sector maximizes profit given prices. The superstar firm maximizes profit given the quality gap, embedded level gap, and number of production lines it has, $(\mathbf{m}, k, n)$, while the fringe firm maximizes profit given prices. The superstar firm chooses the values of vertical z^{Ver} (28), horizontal z^{Hor} (30), and embedded innovation z^{Emb} (28) whereas the fringe firm selects its optimal innovation value z_f (33). The real wage clears the labor market, aggregate consumption and output grow at the same rate, equation (40), and the resource constraint satisfies equation (41).

3. Datasets and Empirical Facts

In this section, I first describe the data sources and measurement details, and then present empirical evidence on how the investment ratio, productivity, markups, and market fluidity vary with firm multiproductness.

3.1. Data Description

Compustat Fundementals and Segment. Compustat Fundamentals provides comprehensive firm-level financial information for publicly listed companies in North America[15] and

[15]Foreign firms such as Toyota and Unilever are included in Compustat North America due to their U.S. listings via American Depositary Receipts (ADRs). Although they adhere to home-country governance, these firms comply with SEC reporting requirements. Excluding them does not affect the paper's main conclusions but reduces the sample size by roughly 25%. For this reason, they are retained in the main analysis. See Figure A1

offers extensive longitudinal coverage. It includes detailed balance sheet items, income statement components, cash flow data, and key financial ratios. An additional advantage is that it enables data to be merged with external datasets through a unique firm identifier.

Compustat also offers two distinct segment datasets[16]: (i) the Historical Segment dataset, which contains buyer–supplier relationships and firm segmentation with long-term coverage[17], and (ii) the Compustat Segment dataset, introduced in 2016, which provides more detailed segment-level information but with limited historical depth. [18] To maximize both coverage and detail, I combine these two segment datasets.[19] To estimate markups, productivity, and the investment ratio across production lines, I merge the Compustat Fundamentals and Segment datasets.[20]

Fludity Dataset. The product market fluidity metric from Hoberg, Phillips, and Prabhala (2014)(HPP) measures the rate at which firms in similar markets change their product or service offerings annually. It is calculated using natural language processing (NLP) on the product descriptions from firms' annual 10-K reports filed with the U.S. Securities and Exchange Commission. This method tracks year-over-year changes in how companies describe their business. A high fluidity score indicates that competitors are adapting rapidly by launching new products, shifting strategies, and entering new markets. Consequently, firms in high-fluidity markets face heightened competitive threats from rivals reconfiguring their offerings and positions.

Forward Citation and Patent Value Dataset. The Kogan, Papanikolaou, Seru, and Stoff-

[16]For a comparative illustration of firm segment classification, see Tables A1.

[17]Under Regulation SFAS No. 131—codified as ASC 280 after 2009—U.S. public firms are required to disclose the identity of any one customer that accounts for more than 10% of its total revenue, along with the nature of the products or services provided to that customer. These mandated disclosures constitute the foundation of the Historical Segment datasets.

[18]In this project, for the Compustat Historical Segment dataset, geographic and operational segments are excluded; only business segments are retained. For the Compustat Segment dataset, only non-missing entries from the Product–Service (`PD-SRVC`) category are included.

[19]When segment information for a firm is available in the Compustat Segment dataset, I prioritize that source. Otherwise, I use data from the Historical Segment dataset.

[20]The merged Compustat dataset contains fewer firms than the Fundamentals database because segment information is unavailable for some firms. In addition, I restrict the sample to firms with positive R&D and SG&A expenditures. This cleaning and merging process does not affect the representativeness of the merged dataset relative to the full Compustat Fundamentals sample; see Figure A4.

man (2017) dataset contains patent-level information, including patent ID, filing and issue dates, firm identifiers, forward citations, and patent value. Forward citations are calculated as the total number of subsequent citations each patent receives, including citations from the firm that owns the patent. The private value of each patent is estimated using stock market reactions around the patent grant date, which captures investors' expectations regarding future profits.

3.2. Measurement

Productivity and Markup Estimation. I estimate firm-level total factor productivity using the approach developed by Gandhi, Navarro, and Rivers (2020)[21]. They propose a non-parametric identification strategy that uses a transformation of the first-order condition for intermediate inputs to isolate flexible-input effects and identify the production function and input elasticities without relying solely on proxy inversion. To estimate firm-level markups, I follow the methodology of De Loecker, Eeckhout, and Unger (2020) and define markups as the ratio of sales to the cost of goods sold (cogs) multiplied by the output elasticity of the variable input, which I obtain from the first-stage production function estimation using Levinsohn and Petrin, 2003.

Investment Ratio. Following Peters and Taylor (2017), I measure two categories of intangible investment. I treat total R&D expenditures as investment in transferable intangibles.[22] On the other hand, 30% of Selling, General, and Administrative (SG&A) expenses, net of R&D, is treated as investment in embedded intangibles. Net SG&A primarily includes employee compensation, advertising, and other expenditures necessary to sustain firm operations. Only a fraction of these expenses are considered investments, as the remainder reflects routine operating costs.

[21]The productivity estimation results are robust to alternative production function estimation methods, including those proposed by Ackerberg, Caves, and Frazer (2015) and Levinsohn and Petrin (2003); see Figure A3. For methodological details, see Appendix A.1

[22]Peters and Taylor (2017) refer to this as knowledge capital, which is termed transferable intangibles in this paper.

3.3. Empirical Facts

Figure 2(A) and Figure 3(A) document an inverse relationship between a firm's number of production lines and its productivity, productivity growth, and markups[23]. This pattern is consistent with a span of control constraint: on average, as firms expand their scope, managerial attention is weakened on each line, reducing efficiency and pricing power.

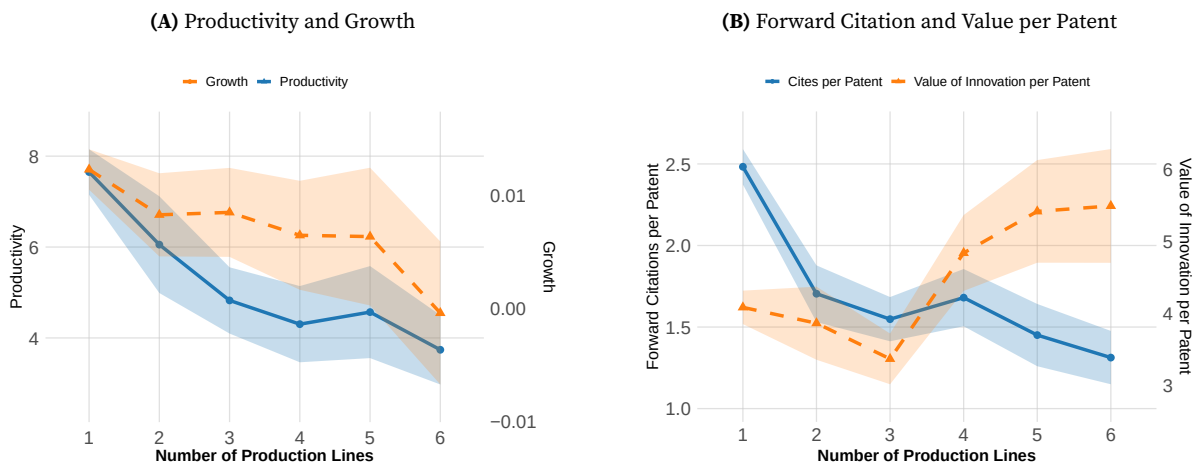

Figure 2. Productivity, Growth & Forward Citation / Value per Patent by Production Line

Note: The sample excludes firms in the utilities and finance sectors, and those with missing or non-positive R&D or SG&A. For forward citations and patent values, observations with missing or non-positive patent values are also excluded. The growth rate is defined as the two-year log change in productivity, $\Delta_2 \ln(\text{prod})_{i,t} = \ln(\text{prod}_{i,t}) - \ln(\text{prod}_{i,t-2})$, averaged over 2005–2019 and winsorized at the 10th and 90th percentiles. Log productivity is measured from the 2019 cross-section and winsorized at the 95th percentile. Forward citations and value per patent are averaged over 2015–2019 and winsorized at the 5th and 95th percentiles.

Figure 2(B) suggests that, on average, forward citations per patent decline as the numbers of production lines increase. This pattern implies a reduction in the social value of patents as firms diversify. In contrast, the private value of innovation per patent appears to rise with additional production lines, moving in the opposite direction from forward citations. The growing divergence between these two measures highlights a misalignment between the private surplus captured by firms and the social value generated by their patent output. Further, Figure 3(B) indicates that the composition of intangible in-

[23]This finding aligns with the pattern in Autor, Dorn, Katz, Patterson, and Van Reenen (2020) in which higher-productivity firms charge higher markups.

vestments is non-uniform and varies systematically with firm scope. Firms with fewer lines tend to prioritize transferable intangibles, whereas firms that expand into multiple products increasingly allocate investment toward embedded intangibles to reinforce competitive advantages in existing lines. Moreover, Figure 3(C) shows that markets dominated by multiproduct firms are less fluid than those dominated by firms with fewer products.[24] A strong negative correlation between market fluidity and the investment ratio suggests that larger embedded intangible investments strengthen incumbency, making market entry and displacement more difficult for rivals. These relationships are formally confirmed by the regression estimates in appendix Table A3, which control for two-way fixed effects. The results show a significant negative relationship between the number of production lines and both productivity and markups, with even stronger negative coefficients for investment ratios and market fluidity. Crucially, appendix Figure A2 demonstrates that these patterns are not explained solely by firm size or age, underscoring their unique link to a firm's expansion strategy.

Further, I apply the local projection method by Jordà (2005) to investigate how single- and multi-segment firms systematically differ in their responses to innovation shocks across productivity, markup, investment ratio, and fluidity.[25] The econometric specification is as follows:

$$\begin{aligned}\Delta_h Y_{i,t} = \alpha_h + \beta_{\text{Multi}}\, S_{it} D_{it} + \beta_{\text{Single}}\, S_{it}\big(1 - D_{it}\big) \\ + \Gamma_h^\top \mathbf{X}_{it-1} + \delta_h Y_{i,t} + \theta_j + \lambda_t + \varepsilon_{ijt}.\end{aligned} \tag{42}$$

Here, $\Delta_h Y_{i,t}$ denotes the h-period change in the outcome Y for firm i, S_{it} is the sum of all innovation shocks to firm i in year t, and D_{it} is an indicator that equals one for multi-segment firms according to Compustat classification (and zero otherwise). The vector $\mathbf{X}_{it-1}$ contains time-varying firm controls, including sales, R&D, and general administrative expenses (SG&A), while Y_{it} controls for the pre-shock level of the outcome. The

[24] See Table A2 for summary statistics.

[25] The innovation shock measure from Kogan, Papanikolaou, Seru, and Stoffman (2017) captures the market reaction to the date a firm's patent was granted.

terms θ_j and λ_t are industry (2-digit NAICS) and year fixed effects, respectively, and ε_{ijt} is the error term. The coefficients β_{Multi} and β_{Single} capture the responses of multi- and single-segment firms, respectively, to the innovation shock.

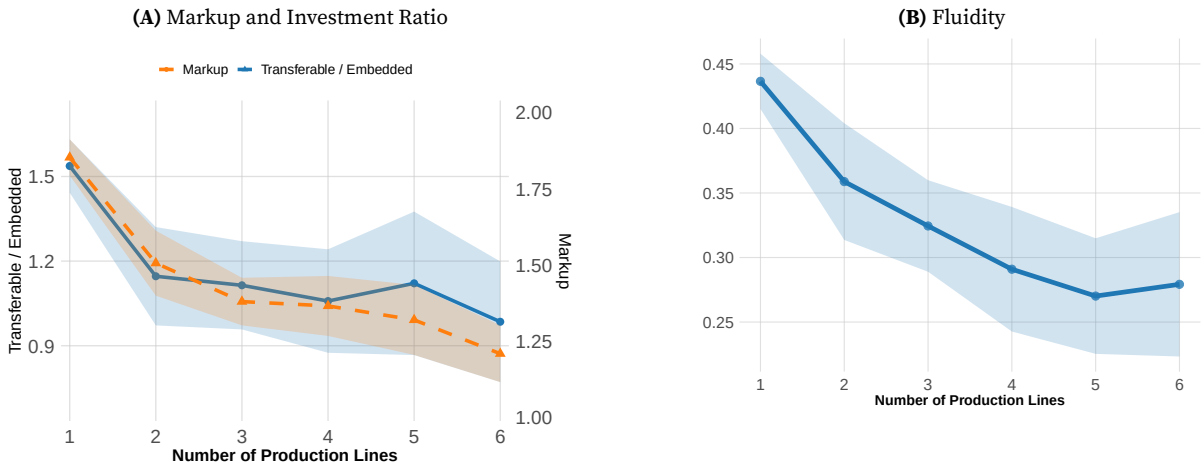

Figure 3. Markup, Investment Ratio, and Fluidity with Production Lines

Note: The sample excludes utilities and finance sectors, and firms with missing or non-positive R&D and SG&A. Markups, the investment ratio, and market fluidity are measured for the 2019 cross-section. The investment ratio is transferable over embedded investment, which is described in Section 3.2. The investment ratio is winsorized at the 95th percentile, the markup at the 90th percentile, and labor market fluidity at the top and bottom 5th percentiles. For calibration purposes, fluidity is normalized using min-max scaling. Each value x was transformed according to $x_{\text{scaled}} = (x - \min(x))/(\max(x) - \min(x))$, mapping all values linearly into the range $[0, 1]$.

Figure 4 shows that after an innovation shock, single-segment firms indicate statistically significant higher investment in transferable intangibles and higher markups. Although the confidence intervals for productivity and fluidity overlap, the point estimates are consistently higher for single-segment firms. This suggests that single-segment firms are more efficient and that their innovations make markets more fluid compared to multi-segment firms.

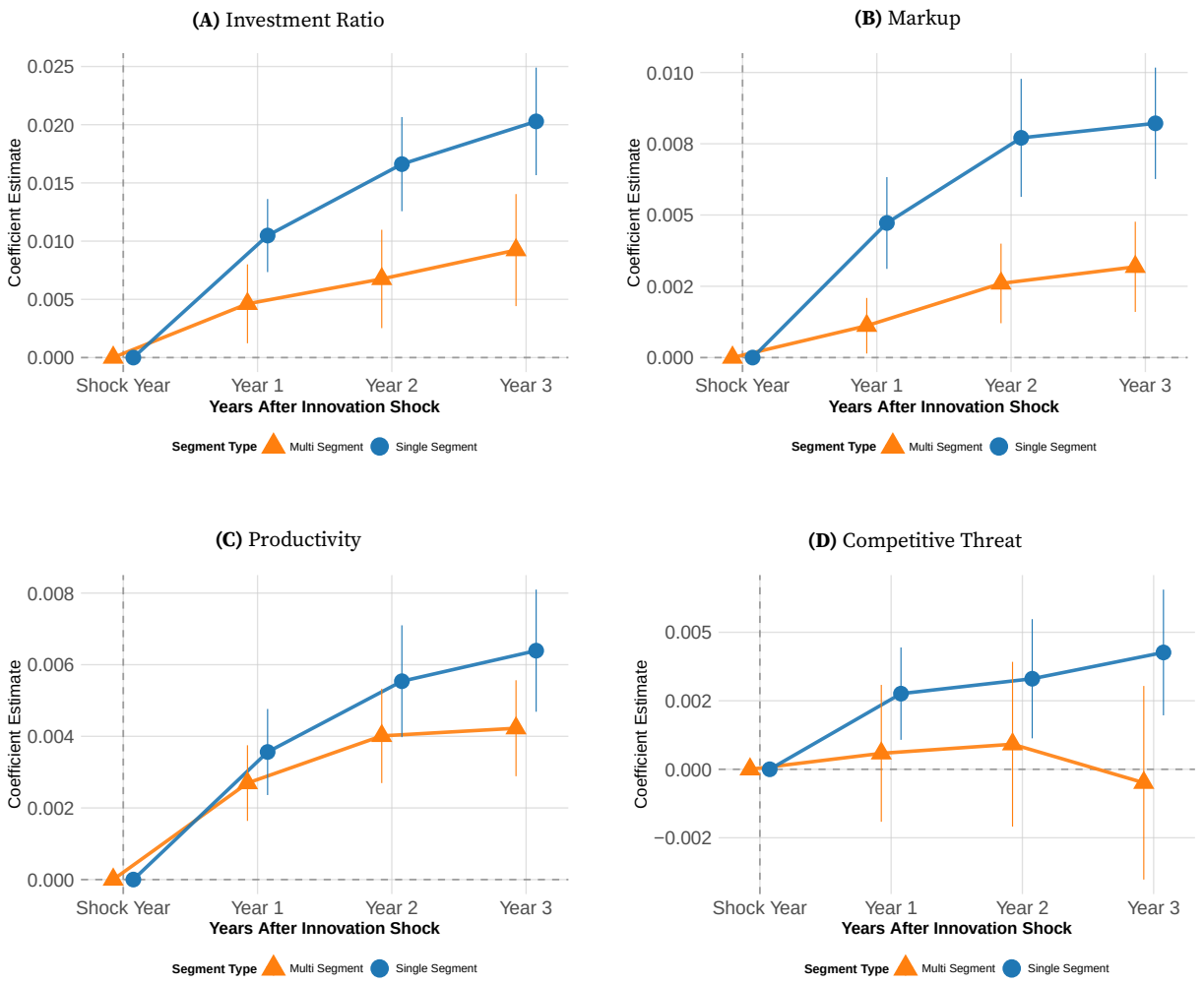

Figure 4. Innovation Shock to Single and Multi-Segment Firms

Note: The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A. All variables are in logarithmic form and cover the period 1990–2019.

4. Quantitative Analysis

In this section, I present calibration parameters to illustrate how the model responds to variations in the investment ratio of transferable to embedded intangibles, and to changes in productivity, growth rate, and markup. Section 5 then examines counterfactual scenarios, first shutting down the effects of embedded intangibles and then considering the span-of-control problem, to assess their impact on the economy. Section 6 subsequently

discusses misallocation and the associated policy implications.

4.1. Empirical counterpart

For each production line n, I aggregate the joint distribution over the quality gap and the embedded-intangible level, described as

$$\mu(n) \;=\; \sum_m \sum_k \mu(m,k,n).$$

Fluidity at production line n is measured as the flow rate of incumbent replacement. It is given by[26]

$$\text{Fluidity}(n) \;=\; \frac{\displaystyle\sum_m \sum_{k'>k} \Big(Z^{\text{Ex}}(m,k',n) \Big)\, \mu(m,k,n) \;+\; \sum_m \sum_k Z^f(m,k,n)\, \mu(m,k,n)}{\mu(n)}. \tag{43}$$

The average markup and the investment ratio at production line n are

$$\sigma(n) \;=\; \frac{\displaystyle\sum_m \sum_k \frac{1-\varepsilon\,\phi(m,k,n)}{(1-\phi(m,k,n))\,\varepsilon}\, \mu(m,k,n)}{\mu(n)}, \tag{44}$$

$$\frac{I^T}{I^{\text{Emb}}}(n) \;=\; \frac{\displaystyle\sum_m \sum_k \frac{I^{\text{Int}}(m,k,n) + I^{\text{Ex}}(m,k,n) + I^f(m,k,n)}{I^{\text{Emb}}(m,k,n)}\, \mu(m,k,n)}{\mu(n)}. \tag{45}$$

The growth and productivity by production line are

$$g(n) \;=\; \frac{\ln\lambda \displaystyle\sum_m \sum_k \Big(z^{\text{Int}}(m,k,n) + p^{\text{Ex}}_{k\geq k'}\, z^{\text{Ex}}(m,k,n) + Z^f(m,k,n) \Big)\, \mu(m,k,n)}{\mu(n)}, \tag{46}$$

$$Q(n) \;=\; e^{\,g(n)}. \tag{47}$$

[26] The first term in the numerator captures external innovations (from higher embedded levels $k' > k$) that displace incumbents in (m,k,n); the second term captures innovation coming from the fringe.

4.2. Calibration and Model Performance

The model is disciplined by 36 empirical moments, comprising 18 targeted and 18 untargeted moments. The calibration relies on 17 parameters, of which 4 are set externally. The time-discount rate is fixed at $\rho = 0.05$, and the quality-improvement step size is set to $\lambda = 1.10$ following Akcigit and Ates (2023). The cost scale and curvature of embedded intangibles are taken from Cavenaile, Celik, Roldan-Blanco, and Tian (2025), with $\gamma^{\text{emb}} = 0.0664$ and $\vartheta^{\text{emb}} = 3.3646$. The remaining 13 parameters,

$$\{\varepsilon, \theta, \alpha_s, \alpha, \gamma, \beta, \xi, \gamma_{\text{Int}}, \gamma_{\text{Ex}}, \gamma_f, \vartheta_{\text{Int}}, \vartheta_{\text{Ex}}, \vartheta_f\},$$

are estimated internally. These parameters govern the key structural features of the model. All parameter values are reported in Table 2.[27]

Figure 5 reports the targeted moments used in the simulated method of moments (SMM) estimation, focusing on markup dynamics, the ratio of transferable-to-embedded intangible investment, and the distribution of firms across production lines. Overall, the model reproduces the principal empirical patterns: it captures both the direction and magnitude of the observed trends. Panel (A) shows that the model tracks the decline in markups as the number of production lines increases, although modest deviations remain at the extremes of the distribution—the model understates markups for single-line firms and slightly overstates them for firms operating many lines. Panel (B) illustrates that the transferable-to-embedded investment ratio is well matched across production-line categories, with simulated moments closely following the empirical shape. Panel (C) demonstrates a strong fit for the firm distribution, in which simulated shares align closely with observed data. Taken together, these results suggest that the model is well disciplined by the targeted moments and captures key margins of firm behavior, with only minor discrepancies concentrated in the tails of the markup profile.

Further, Figure C1 evaluates the model's performance on untargeted moments not used

[27] Appendix C.2 provides details of the solution algorithm and the simulated method of moments.

Table 2. Parameter Values

Parameter	Description	Value
	———————*External Calibration*———————	
ρ	Discount rate	0.05
λ	Transferable innovation step size	1.0100
γ^{Emb}	Cost scale of embedded innovation	0.0664
ϑ^{Emb}	Curvature of embedded innovation	3.3646
	———————*Internal Calibration*———————	
ε	CES parameter	0.7747
θ	Embedded innovation step size	1.0100
α	Curvature of managerial productivity	0.5980
γ	Scale of managerial productivity	0.4001
β	Curvature of brand value	0.0661
ξ	Share of brand value on embedded intangible	0.4351
γ^{Int}	Cost scale of internal innovation	2.8571
γ^{Ex}	Cost scale of horizontal innovation	0.4001
γ^{f}	Cost scale of fringe	5.1777
ϑ^{Int}	Curvature of internal innovation	15.3154
ϑ^{Ex}	Curvature of horizontal innovation	5.4183
ϑ^{f}	Curvature of fringe	7.9917

Note: The upper limit for the number of production lines $\bar{n}$ is set to 6, and the upper bounds for $\bar{m}$ and $\bar{k}$ are set to 9.

in the SMM, including productivity, aggregate growth rates, and measures of fluidity across production-line categories. Panel (A) indicates that the model captures the declining pattern of productivity as production-line count rises, with only small departures from the data at intermediate values. Panel (B) reveals systematic differences in the aggregate growth rate: the model tends to underestimate actual growth for firms with one to three production lines and slightly overshoots growth at the upper end of the distribution. These discrepancies imply that, while the model reproduces the overall downward growth trend, it misses some non-monotonic features present in the data. Panel (C) shows that the model generally undershoots empirical fluidity measures across most production-line categories, with the notable exception of the sixth line, where simulated fluidity converges more closely to the observed value.

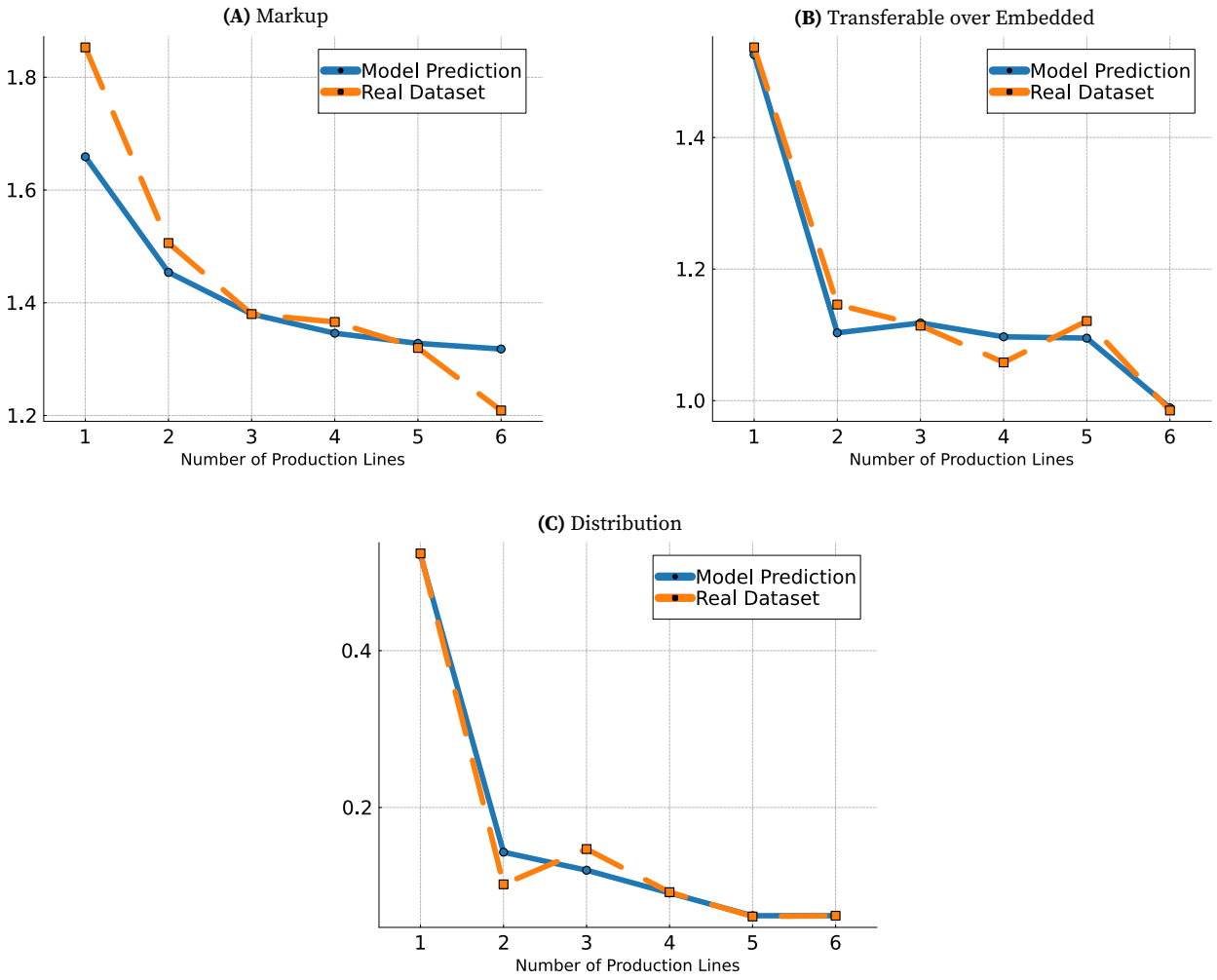

Figure 5. Targeted Moments: Markup, Investment Ratio, and Fluidity by Production Lines

Note: The orange line represents the dataset values, while the blue line shows the model simulation results along the balanced growth path. The horizontal axis corresponds to the production line dimension.

4.3. Firm Innovation Direction Decision

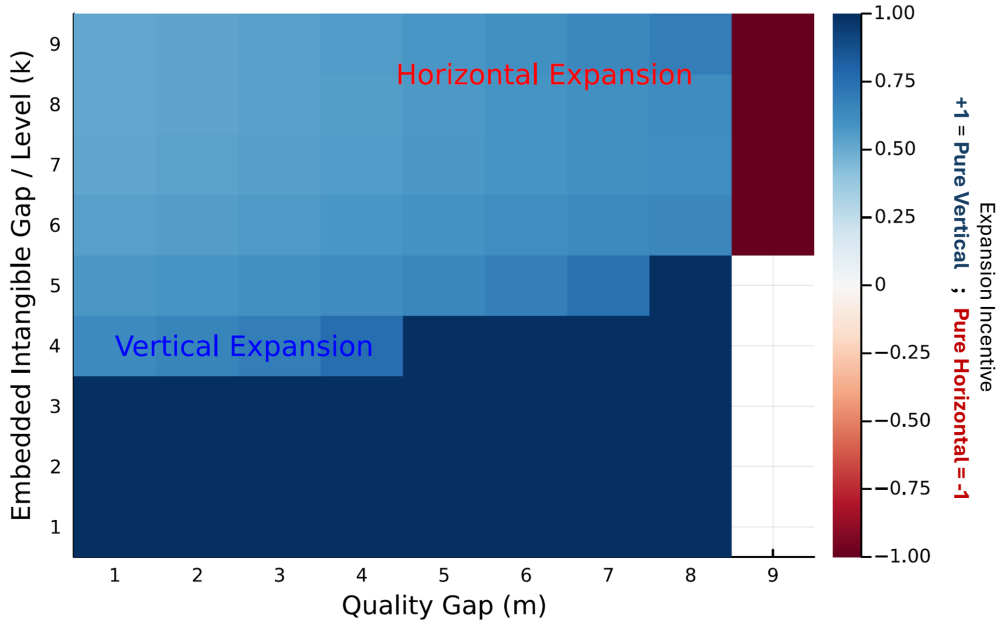

Figure 6. Superstar Firms Innovation Direction (Vertical vs Horizontal)

Figure 6 illustrates the firm's optimal innovation direction as a function of the quality gap, m, and the embedded intangible gap, k. The horizontal axis captures the firm's position on the quality ladder relative to competitors, while the vertical axis measures the embedded intangibles levels. The color scale reports the normalized innovation incentive: darker blue regions indicate a stronger incentive for vertical innovation, whereas darker red regions indicate a stronger incentive for horizontal innovation. Intermediate shades reflect regions in which the two strategies yield comparable payoffs.

The figure reveals a clear pattern. When the embedded intangible gap k is low, the firm optimally concentrates on vertical innovation. In this region, embedded assets are insufficient to generate substantial cross-sector economies, so the marginal return to improving existing products dominates the return to expanding scope. As k increases, however, embedded intangibles amplify cross-sector synergies and enhance the value of operating in multiple industries. Beyond a threshold level of k, horizontal expansion becomes the preferred strategy, even if the quality gap m remains moderate.

4.4. Demand and Supply Effect of Embedded Intangibles

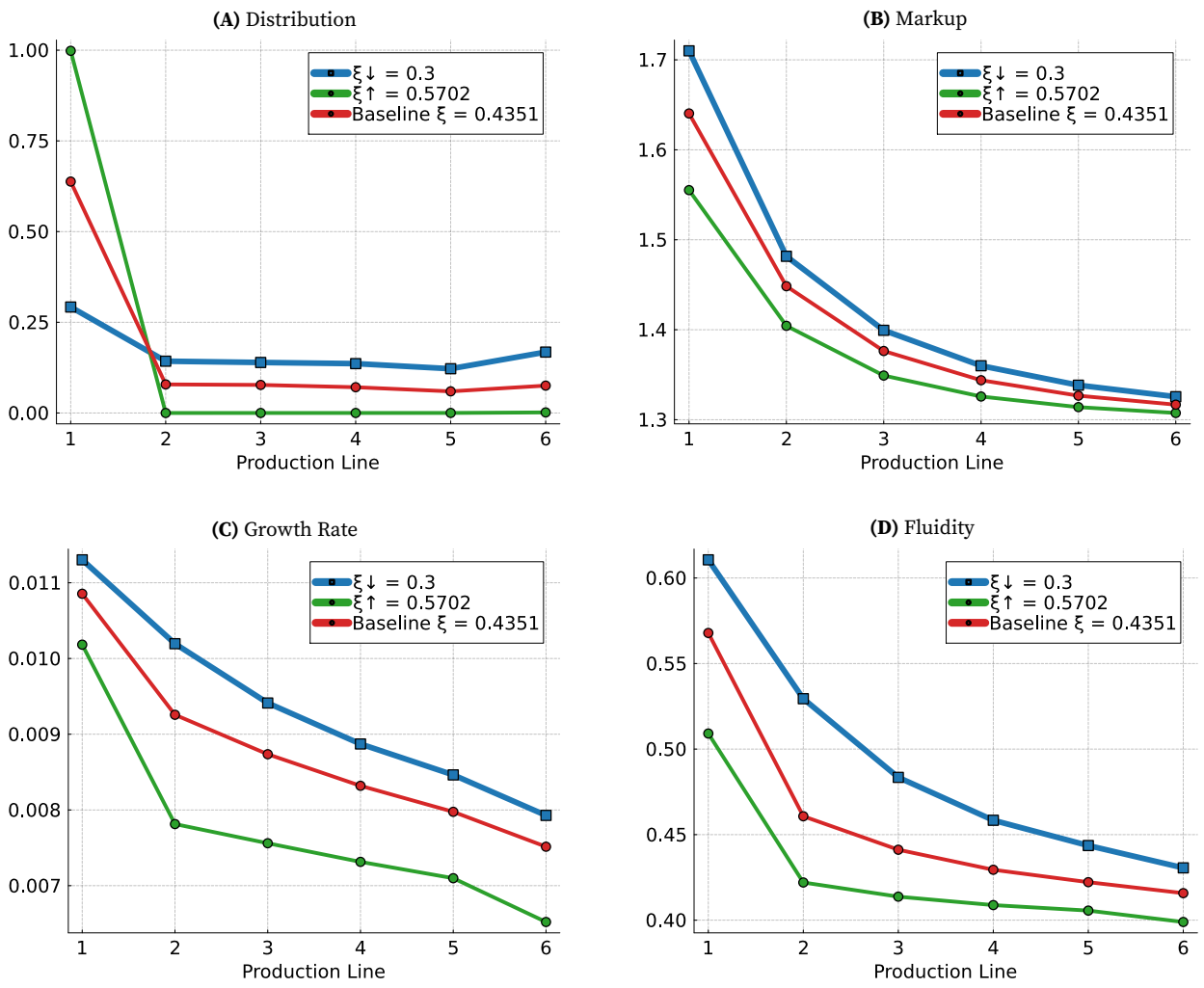

Figure 7. Impact of ξ on Distribution, Markup, Growth, and Fluidity

Note: The green line represents the internally calibrated optimal value of ξ, while the blue line shows an upward shift in ξ and the orange line shows a downward shift. The horizontal axis corresponds to the production line dimension.

In the model, parameter ξ governs the share of brand value (demand side), while $1 - \xi$ corresponds to organizational capital (supply side). A higher ξ implies that brand value dominates (green line), whereas a lower ξ (blue line) indicates a greater role for organizational capital. Figure 7 shows that firms derive greater benefits from organizational capital than from brand value when expanding their scope. The primary reason for this is that organizational capital can directly offset some of the managerial difficulties asso-

ciated with expansion, thereby promoting growth across production lines.

In contrast, brand value does not provide this offsetting capability. Consequently, when the share of brand value (ξ) is high, firms tend to stop expanding and predominantly operate a single production line. The associated decrease in markups may seem counterintuitive at first. However, this decline is primarily driven by reductions in organizational capital, which diminishes managerial productivity gains. This negative supply-side effect outweighs any positive demand-side effects from brand value. [28]Moreover, when ξ is high, superstar firms have no incentive to expand horizontally; therefore, they concentrate on a single line. This accumulation creates a significant barrier to entry, as new entrants must match this high level of intangibles to compete successfully even on a single line. As a result, both market fluidity and the aggregate growth rate decrease. All these effects are reversed when parameter ξ is lower. In this case, the larger share of organizational capital provides firms a greater advantage, facilitating expansion and improving the overall dynamics of creative destruction and growth. [29]

5. Counterfactual Analysis

In this section, I conduct a series of counterfactual analyses to isolate the mechanisms driving the results. First, I deactivate the span of control constraint. Second, I shut down the accumulation of embedded intangibles. Finally, I shut down both mechanisms jointly. This sequence allows me to quantify how each feature—and their interaction—affects key outcomes: markups, the firm size distribution, market fluidity, and the aggregate growth rate of the economy.

Shutting Down the Span of Control Constraint. To evaluate the impact of the span-of-

[28]In the calibration, the curvature of brand value is relatively low compared with that of managerial productivity, which causes supply-side effects to dominate demand-side effects. The parameter γ, which captures the benefit scale of managerial quality, further reinforces this dominance by amplifying supply-side responses relative to demand.

[29]This increase occurs relative to the composition of brand value versus organizational capital. By contrast, Section 6 shows that increasing embedded intangibles leads to substantial inefficiency, regardless of brand value or organizational capital.

control constraint, I conduct a counterfactual analysis by setting the parameter α_s to zero[30]. This removes the mechanism that causes managerial productivity to decline with the number of production lines. The results are presented in Figure 8, where the counterfactual scenario is plotted in green and the baseline calibration in red.

The counterfactual generates a rightward shift in the firm size distribution, with firms operating more product lines relative to the baseline. This expansion follows directly from the removal of span-of-control constraints: in the absence of diminishing managerial returns, firms can add product lines without increasing marginal costs. As a result, markups now increase with the number of product lines, reversing the baseline pattern in which diversification reduced markups due to rising marginal costs from span-of-control frictions. Under Bertrand competition, a firm's markup in each product line is constrained by both its own marginal cost and the marginal costs of its competitors. In the baseline, span-of-control limitations increase the firm's marginal costs as it diversifies, reducing markups. Eliminating this channel keeps marginal costs low across all lines, allowing firms to sustain higher markups as they expand.

In contrast, relaxation of managerial constraints reduces market fluidity. As firms grow larger and manage more product lines, they face stronger incentives to invest in embedded intangibles due to increasing returns to scale. The resulting accumulation of intangible capital raises entry barriers and lowers overall market fluidity. Despite the decline in market fluidity, the aggregate growth rate rises. This counterintuitive result stems from a shift in the source of growth: while creative destruction diminishes, innovation by incumbent superstar firms increases. Relaxation of span-of-control constraints directly benefits multiproduct firms, which were previously the most constrained. Consequently, their enhanced ability to expand and innovate boosts firm-level growth rates, increasing aggregate growth relative to the baseline.

Shutting Down Embedded Intangibles. Next, I examine the effects of shutting down embedded intangibles by setting $\bar{k} = 1$. In this counterfactual, superstar firms can no longer

[30]The parameter α_s is highly sensitive; setting it directly to zero prevents the model from converging and producing results. Therefore, when $\alpha_s = 0$, $\bar{k}$ is set to 6 rather than 9 in the baseline calibration.

benefit from accumulating brand value or organizational capital.

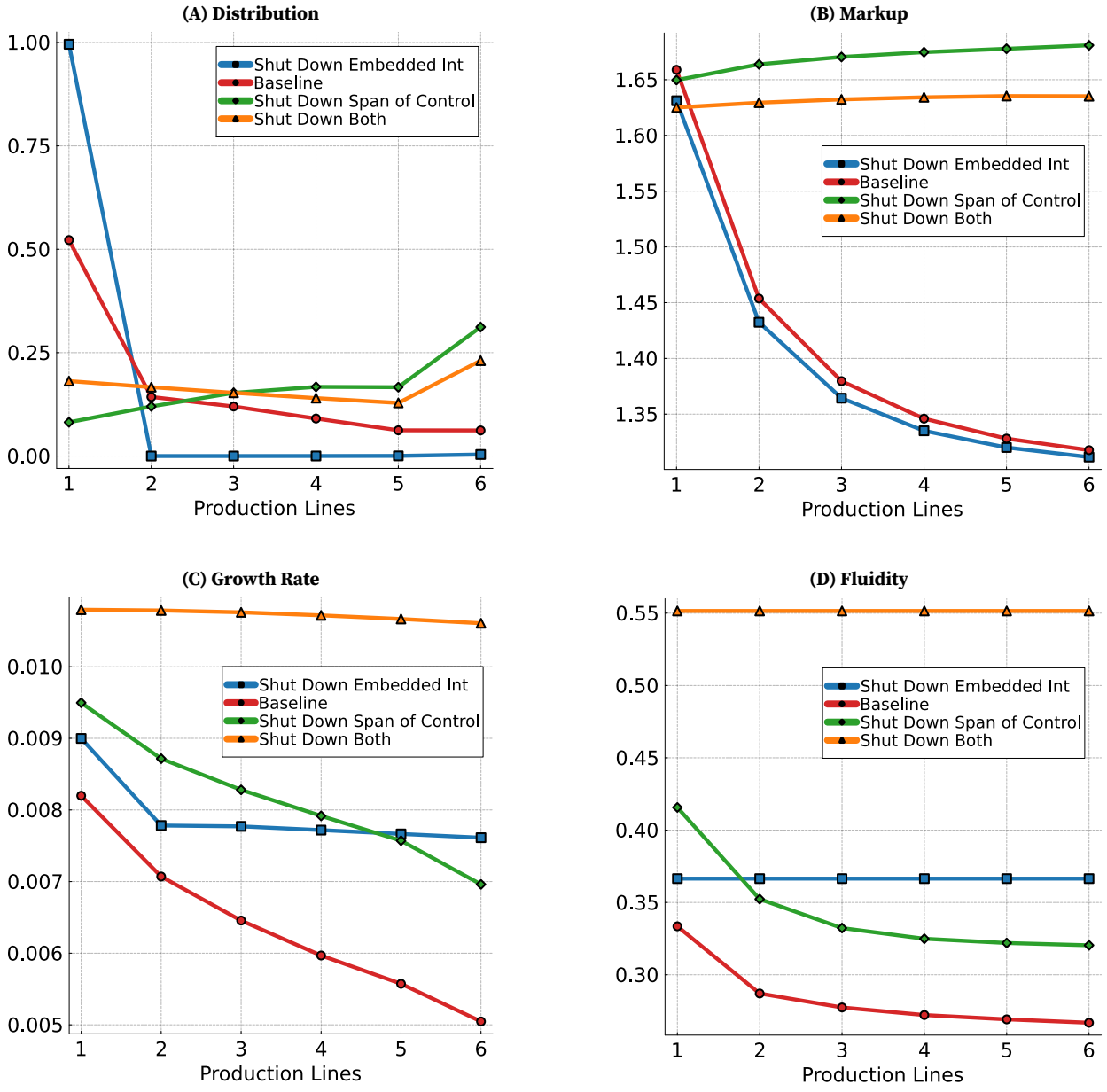

Figure 8. Counterfactual Analysis Under Different Cases

Note: The red line shows the baseline calibration; the green line shows the case with span-of-control constraint shut down ($\alpha_s = 0$); the blue line shows the case with embedded intangibles shut down ($\bar{k} = 1$); and the orange line shows the case with both the span-of-control constraint and embedded intangibles shut down ($\alpha_s = 0$ and $\bar{k} = 1$).

As shown in Figure 8 (blue line), this leads to a leftward shift in the firm size distribution, which becomes concentrated on a single production line. The mechanism driving this shift is that expanding to additional lines remains costly due to the persistent span-

of-control constraint, and firms can no longer offset these costs by improving managerial quality through embedded intangibles. Markups also decline, as firms lose the pricing advantage provided by embedded intangibles relative to rivals. However, this reduction is modest, indicating that, while embedded intangibles contribute to markup growth, their overall quantitative impact is limited.

The aggregate growth rate rises, primarily driven by a significant increase in market fluidity. Eliminating the accumulation of embedded intangibles removes entry barriers for other superstar firms, allowing them to enter production lines more easily without facing entrenched incumbents. This enhanced entry raises the rate of creative destruction, and the resulting higher market fluidity directly contributes to the increase in aggregate growth. These results highlight the important role of embedded intangibles in shaping market dynamics and economic growth.

Both the removal of span-of-control constraints and the elimination of embedded intangibles raise the aggregate growth rate; however, the sources and magnitudes of these increases differ. In the span-of-control counterfactual, the growth boost is driven primarily by incumbents. In contrast, when embedded intangibles are eliminated, the growth increase is driven by innovations from other superstar firms on incumbent lines.

Shutting Down Both of Mechanisms. Shutting down both mechanisms simultaneously allows multiproduct firms to expand horizontally, which shifts the firm size distribution rightward. However, the magnitude of this shift is significantly limited compared to the scenario in which only the span-of-control constraint is removed. The primary reason is that, in this combined counterfactual, superstar firms can no longer leverage the increasing returns to scale afforded by embedded intangibles to facilitate their expansion. For the same reason, markups increase with the number of production lines, but this increase is more muted than in the counterfactual involving only the span-of-control constraint. Without embedded intangibles, firms lack one of the tools to amplify their pricing power as they grow.

On the other hand, market fluidity and the aggregate growth rate are higher in this combined counterfactual than in either single shutdown case. This elevated growth stems

from a dual source: intensified innovation by incumbents and heightened innovation by potential superstar firms in other industries, who find it easier to enter the market and to contest existing production lines due to lower barriers arising from the absence of embedded intangibles.

To summarize, all three counterfactuals demonstrate that both span-of-control constraints and embedded intangibles are necessary to replicate the empirical patterns shown in Section 3.3. Omitting either component results in deviations from the observed facts: markups no longer decrease with scope, firm size distribution becomes concentrated on a single production line, and the declining trends in growth and market fluidity across production lines are not reproduced. Therefore, these two mechanisms together provide the minimal conditions required to capture the key empirical regularities.

6. Policy Implication and Misallocation

This section first quantifies the effects of resource misallocation on aggregate output and economic growth. Based on these insights, I analyze policy tools to examine how these frictions affect welfare and how they can be mitigated.

6.1. Misallocation

Misallocation in the model operates through two distinct channels. The first stems from markup dispersion across firms. The second arises because a potential superstar entrant cannot capture a market unless its embedded intangible level is at least as high as the incumbent's, creating a barrier to entry. To quantify the effect of markup dispersion, I adapt the method by Peters (2020) and decompose aggregate output Y into four components: the contribution from quality improvements (Q), the contribution from embedded intangible capital (E), the misallocation due to markup dispersion (M), and a leftover term (S)[31],

$$Y_t = Q_t \times E_t \times M_t \times S_t \tag{48}$$

[31]For details see Appendix B.6

$$\text{where} \quad E_t = \exp\left(\sum_m \sum_k \sum_n \frac{1}{\varepsilon} \ln\left[(\xi\theta)^{k\times\beta}\left((1-\xi)\theta\right)^{k\times\alpha}\right)^{\varepsilon}\right] \mu(m,k,n)\right),$$

$$M_t = \frac{\exp\left(\sum_m \sum_k \sum_n \ln\left[\frac{\phi(m,k,n)}{\sigma(m,k,n)} + (1-\phi(m,k,n)\right] \mu(m,k,n)\right)}{\sum_m \sum_k \sum_n \left(\frac{\phi(m,k,n)}{\sigma(m,k,n)} + (1-\phi(m,k,n)\right) \mu(m,k,n)}, \quad \text{and}$$

$$S_t = \exp\left(\sum_m \sum_k \sum_n \ln\left[\frac{\left(\frac{1}{\gamma n^{\alpha_s}} \frac{\phi(m,k,n)}{\sigma(m,k,n)}\right)^{\varepsilon} + \frac{1}{(\xi\theta^k)^{\beta}}\left(\frac{\lambda^{-m}}{((1-\xi)\theta^k)^{\alpha}}(1-\phi(m,k,n))\right)^{\varepsilon}}{\left(\frac{\phi(m,k,n)}{\sigma(m,k,n)} + (1-\phi(m,k,n))\right)^{\varepsilon}}\right]^{\frac{1}{\varepsilon}} \mu(m,k,n)\right).$$

Defining M as the ratio of the geometric mean to the arithmetic mean, the key misallocation term endogenously evolves with markup dispersion. [32] The results, presented in Table 3, show that although there is substantial markup dispersion across production lines, the aggregate dispersion effect on misallocation is relatively small. As a result, the misallocation arising purely from markup dispersion is minimal, accounting for only about 0.5% of output.

To analyze the second type of misallocation, I conduct a counterfactual experiment that eliminates entry barriers. In this scenario, any superstar firm that successfully makes a horizontal innovation can immediately enter a new production line and become the incumbent without facing the embedded intangible constraint. The results are striking: aggregate output more than triples. This dramatic gain is primarily driven by a large increase in the quality improvement component (Q). The contribution from embedded intangibles (E) slightly decreases, a likely result of the policy boosting investment in transferable technology (horizontal innovation) and making embedded investment less critical for market entry. Furthermore, the markup dispersion term (M) approaches one, and the residual term (S) increases. These results suggest that policy attention should focus on lowering entry barriers. Enabling frictionless reallocation following horizontal innovations substantially raises aggregate output, primarily through higher quality growth, and

[32] The markup dispersion term M differs slightly from Peters (2020). Even if markups were constant across superstar firms, dispersion between superstars and fringe firms would still exist, governed by the market share ϕ of superstar firms.

therefore can materially reduce misallocation in the economy.

Table 3. Misallocation

	Y	Q	E	M	S	g
Base Scenario	2.512	1.923	0.691	0.995	1.900	0.007
Shut Down Entry Barrier	7.148	4.588	0.673	1.000	2.314	0.017

6.2. Policy Implication

Based on the frictions discussed in Section 6.1, I analyze three distinct tax regimes. First, I implement a size-dependent tax on profit starting at 10% and gradually increasing to 12.5%. Second, I consider a flat tax of 11.3% specifically on investments in embedded intangibles and horizontal expansion. Third, I examine the joint implementation of both tax policies simultaneously. In all cases, the government collects the tax revenue and redistributes it to households through lump-sum transfers. To evaluate the welfare effects of these policies, I employ a consumption-equivalent measure[33] This metric quantifies the permanent percentage change in consumption that would make households indifferent between the baseline economic path and the policy-induced path. The welfare results show substantial variation across policies. The size-dependent tax generates a welfare increase of 10.915%, accompanied by corresponding rises in consumption and output. In contrast, taxes targeting embedded and expansion investments yield more modest gains of 1.745% and 2.176%, respectively. The size-dependent tax's superior performance operates through two main channels. First, it reduces market segmentation by making firms less exposed to diminishing returns from managerial span of control. Second, it attenuates incentives to over-invest in embedded intangibles by reducing cross-product synergies, ultimately lowering barriers to entry. The joint implementation of all taxes demonstrates significant positive synergies. The total welfare increase of 16.237% exceeds the sum of individual contributions, indicating that the policies work together to mitigate economic distortions and enhance aggregate efficiency relative to the baseline economy.

[33]See Appendix B.7 for details on the consumption-equivalence welfare calculation.

Table 4. Welfare and Output Effects at Varying Tax Levels

	Tax Type			
	Size-Dependent Tax [0.1, 0.105, 0.11, 0.115, 0.12, 0.125]	Embedded Inv Tax 0.113	Expanding Inv Tax 0.113	All Three Combined
Δ Welfare (%)	10.915	1.745	2.176	16.237
ΔC (%)	10.542	1.602	2.093	15.549
ΔY (%)	8.517	1.566	2.084	13.255

7. Conclusion

This paper proposes that, as firms diversify across segments, their markups, productivity, and their ratio of transferable to firm-specific intangible investment decline. To explain this, I develop a unified endogenous growth model in which the strategic allocation of innovative effort emerges directly from firms' expansion decisions. Horizontal expansion introduces span-of-control frictions, which weaken managerial attention and reduce per-segment profitability. In response, multi-segment firms do not simply innovate less; they innovate differently. They strategically reallocate investment away from transferable intangibles, which generate social spillovers and fuel creative destruction, and toward embedded intangibles, which provide private firm-specific advantages. This shift allows firms to exploit cross-product synergies and higher entry barriers, thereby privatizing returns at a significant social cost: it reduces market fluidity and depresses the long-run engine of growth, which is quality-improving innovation.

Quantitative analysis reveals that this mechanism constitutes a major source of economic misallocation. Crucially, I find that the entry barriers created by embedded intangibles are far more consequential for welfare than pure markup dispersion. This finding underscores the need for policies to address this specific distortion. Counterfactual experiments suggest that well-designed interventions, such as a size-dependent tax on profits or a targeted tax on embedded investment, can effectively rebalance private incentives with social goals. By discouraging excessive, friction-inducing diversification

and encouraging R&D, such policies can help realign the trade-off between diversification and growth to foster a more dynamic and productive economy.

This framework establishes a foundation for several routes of future research. First, the evolution of firm scope over the lifecycle would permit a richer analysis of dynamic market segmentation strategies and their long-term consequences for the direction and pace of innovation. Second, a formal welfare analysis comparing a decentralized equilibrium to a social planner's solution is a critical next step. Such a comparison would allow for the precise derivation of optimal policy instruments to counteract the inefficiencies and knowledge-spillover frictions engendered by diversification. Third, the model generates testable empirical predictions regarding how firms endogenously respond to span-of-control constraints. Micro-econometric work could investigate the relative efficacy of adaptive strategies such as organizational redesign, investments in information technology, and human capital accumulation. Finally, a critical question is whether embedded intangible capital slows idea diffusion by limiting spillovers. Answering this with richer microdata and dynamic structural methods will be key to guiding policies that balance the private gains from diversification with the broader social returns to innovation and competition.

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Appendices

A. Empirical Appendix

A.1. Dataset and Measurement Details

Table A1. Example Firms Segment in Compustat Segment Dataset

Company	Segments
TOYOTA MOTOR CORP	Financial Services Automotive All Other
PROCTER & GAMBLE CO	Health Care Grooming Corporate Beauty Baby, Feminine & Family Care Fabric & Home Care
TESLA INC	Energy Generation & Storage Automotive

Measurement Details. The sample covers 1990–2019. The finance and utilities sectors are excluded from all analyses, and estimations are conducted separately for each 2-digit NAICS industry. I employ the approach of Gandhi, Navarro, and Rivers (2020)(GNR) to estimate firm-level total factor productivity. The gross-output production function is specified as

$$Y_{it} = F(L_{it}, K_{it}, M_{it}) + \omega_{it} + \epsilon_{it}, \tag{49}$$

where Y_{it} is (log) gross output (`sale`), L_{it} is labor (`emp`), K_{it} is capital (`ppegt`), and M_{it} denotes flexible (intermediate) inputs (`cogs`). The term ω_{it} denotes an unobserved firm productivity shock that is observed by firms when choosing inputs, and ϵ_{it} is an iid error term.

A central identification challenge is that the presence of flexible inputs creates a non-identification problem for nonparametric gross-output production functions when stan-

dard proxy-variable approaches are applied, because flexible-input choices reflect contemporaneous productivity. GNR resolve this by exploiting a transformation of the firm's short-run first-order condition for intermediates to obtain cross-equation restrictions that isolate the flexible-input contribution and thus permit nonparametric identification of the production function and input elasticities.

Empirical proxy (material share): Define the intermediate share

$$s_{it} \equiv \frac{\texttt{COGS}_{it}}{\texttt{Sales}_{it}},$$

with both numerator and denominator deflated by `cpi`. GNR show that s_{it} is the empirical moment implied by the transformed FOC and can be used to recover the flexible-input elasticity nonparametrically[34].

First stage (nonparametric share regression): I apply the GNR transformation of the FOC for intermediates and estimate the resulting relation between s_{it} and the observable state variables nonparametrically. This yields an observation-level flexible-input elasticity $\hat{\beta}_{m,it}$

Second stage (fixed-input elasticities and TFP): With $\hat{\beta}_{m,it}$ in hand, the second stage identifies the remaining input elasticities $\hat{\beta}_l, \hat{\beta}_k$ using the cross-equation restrictions and standard Markov term $\omega_{it} = g(\omega_{it-1}) + \xi_{it}$. Construct firm TFP as the residual

$$\hat{\omega}_{it} = Y_{it} - \hat{\beta}_l L_{it} - \hat{\beta}_k K_{it} - \hat{\beta}_{m,it} M_{it}.$$

[34]Estimating using methodology by Ackerberg, Caves, and Frazer (2015) and Levinsohn and Petrin (2003) methodology, I replace the proxy variable "revenue share" with "capital expenditure" (`capx`).

A.2. Additional Figures and Empirical Results

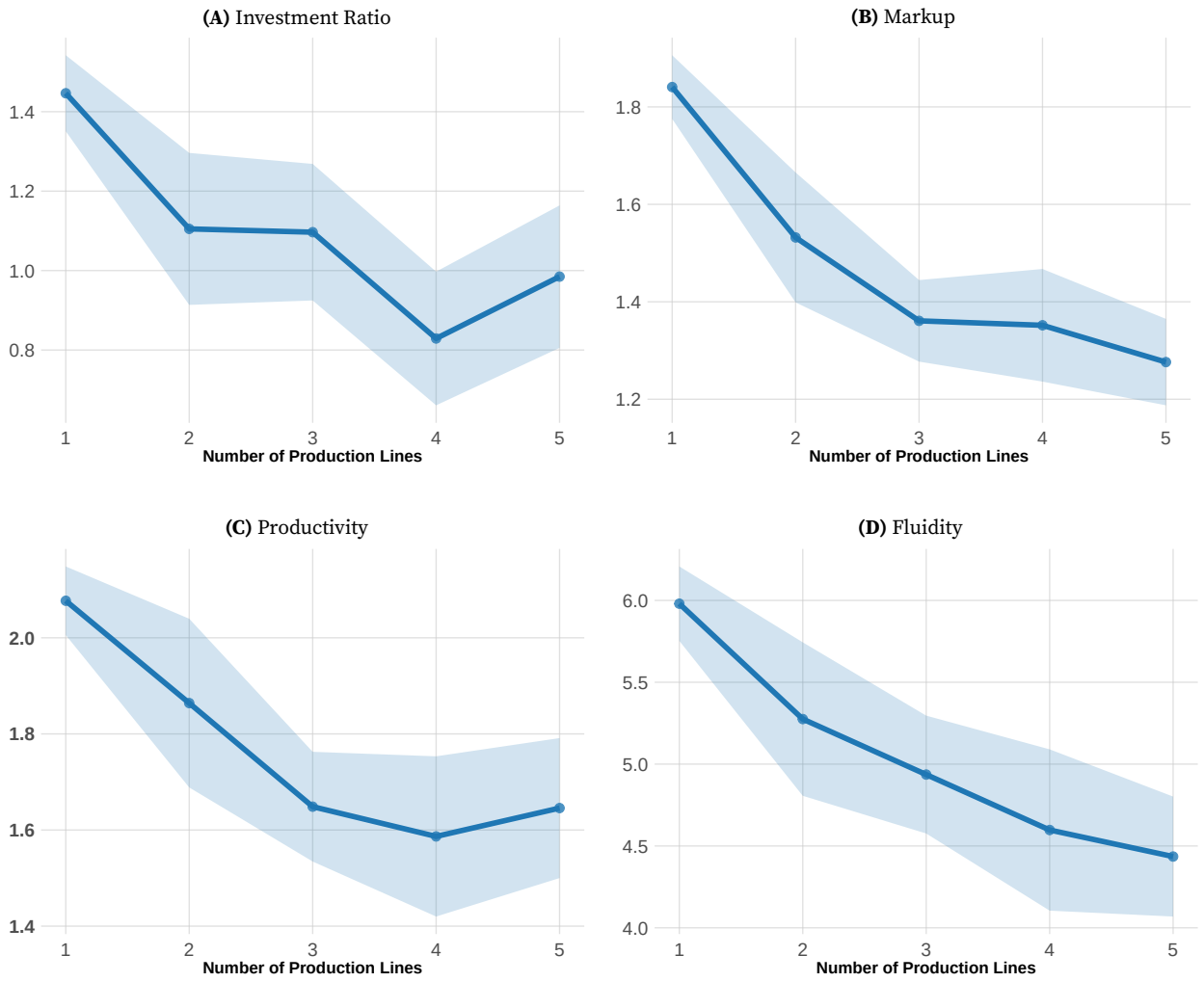

Figure A1. US Firms; Markup, Productivity, Investment Ratio and Fluidity by Production Lines

Note: The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A. Variables are measured for the 2019 cross-section and winsorized at the 95th percentile.

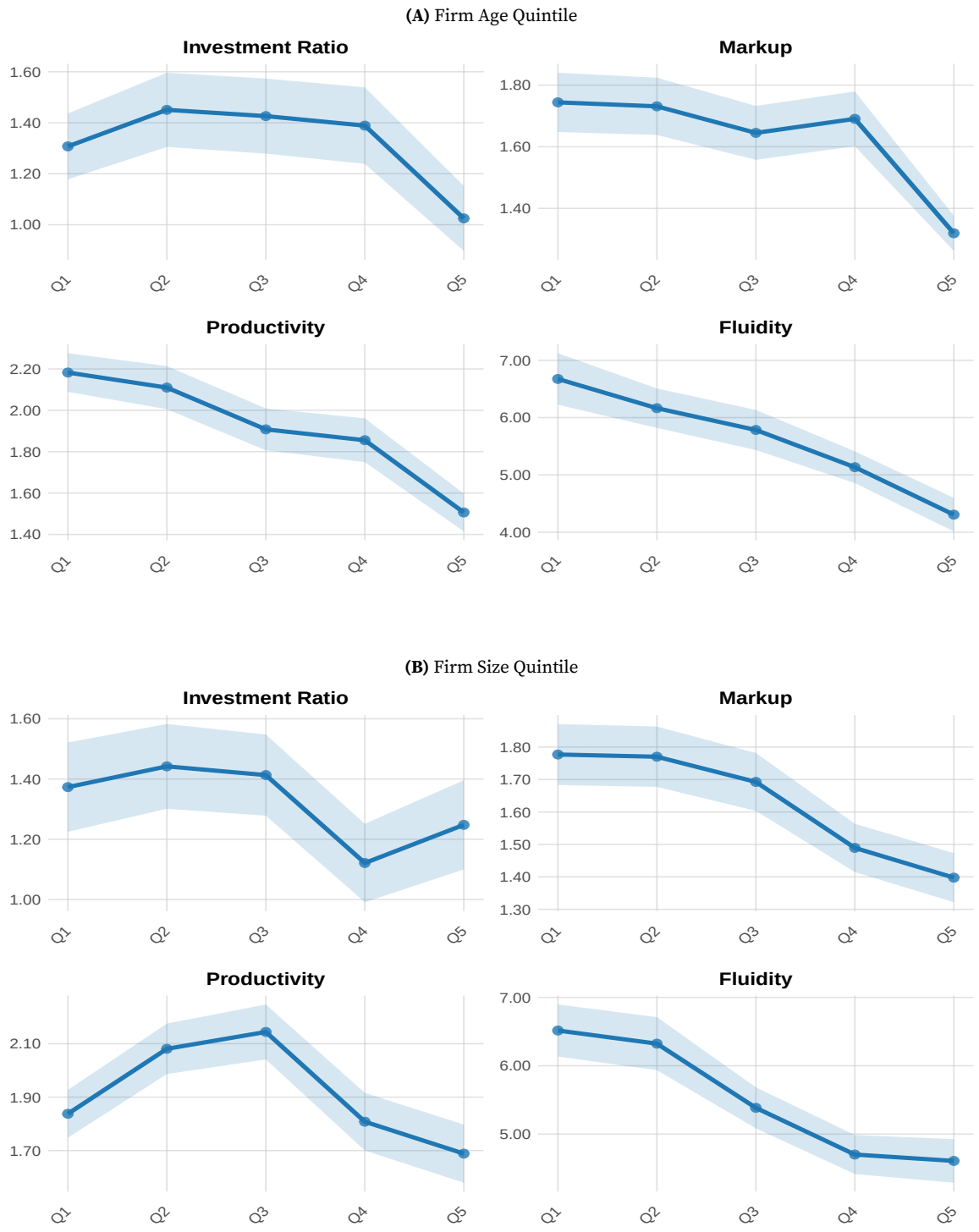

Figure A2. Firm Age and Size

Note: The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A. Markups, the investment ratio, and market fluidity are measured for the 2019 cross-section. All variables are winsorized at the 95th percentile. Firm size is measured by total employment (EMP), and firm age is measured by years since IPO. Firms are grouped into quantiles; the plotted values represent the average within each quantile.

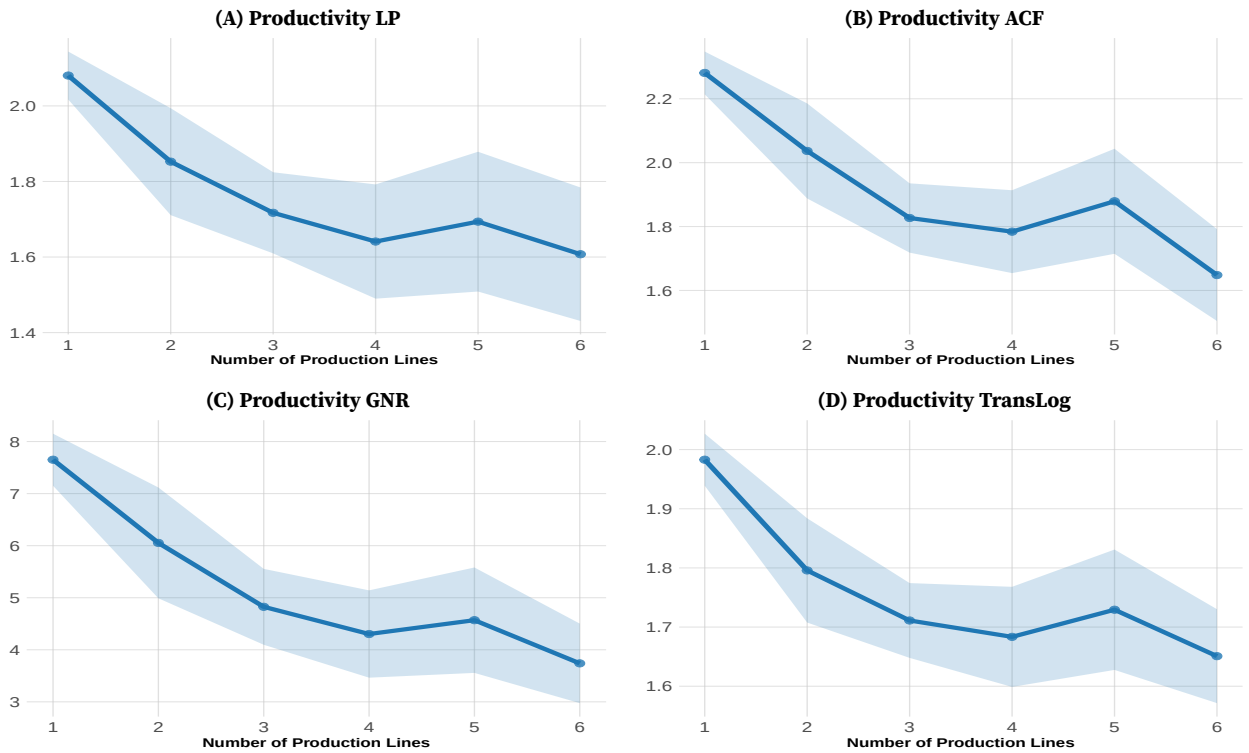

Figure A3. Productivity Measures with Different Methods

Note: The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A. All variables are winsorized at the 95th percentile. Productivities are measured for the 2019 cross-section. LP: Levinsohn and Petrin (2003); ACF: Ackerberg, Caves, and Frazer (2015); GNR: Gandhi, Navarro, and Rivers (2020)

Table A2. Summary Statistics

Variable	Summary Statistics						
	Mean	SD	Median	P10	P25	P75	P90
Investment ratio	1.300	1.310	0.873	0.129	0.317	1.760	3.290
R&D to Sales ratio	0.143	0.187	0.0715	0.006 93	0.0202	0.185	0.364
Markup	1.630	0.830	1.320	0.870	1.020	1.990	3.430
Productivity	1.910	0.972	1.730	0.754	1.010	2.770	3.300
Log sales	13.100	2.570	13.200	9.840	11.400	14.900	16.400
Log total assets	13.600	2.520	13.600	10.300	11.800	15.300	16.800
Log employees	7.490	2.310	7.530	4.380	5.820	9.130	10.500
Number of unique firms				1,711			

This table reports summary statistics for firm characteristics and the main variables used in the paper. The investment ratio of Transferable over Embedded is defined in Section 3. Investment ratio, R&D-to-sales, and productivity are winsorized at the 95th percentile, while markup is winsorized at the 90th percentile. All other variables are presented in logarithmic form.

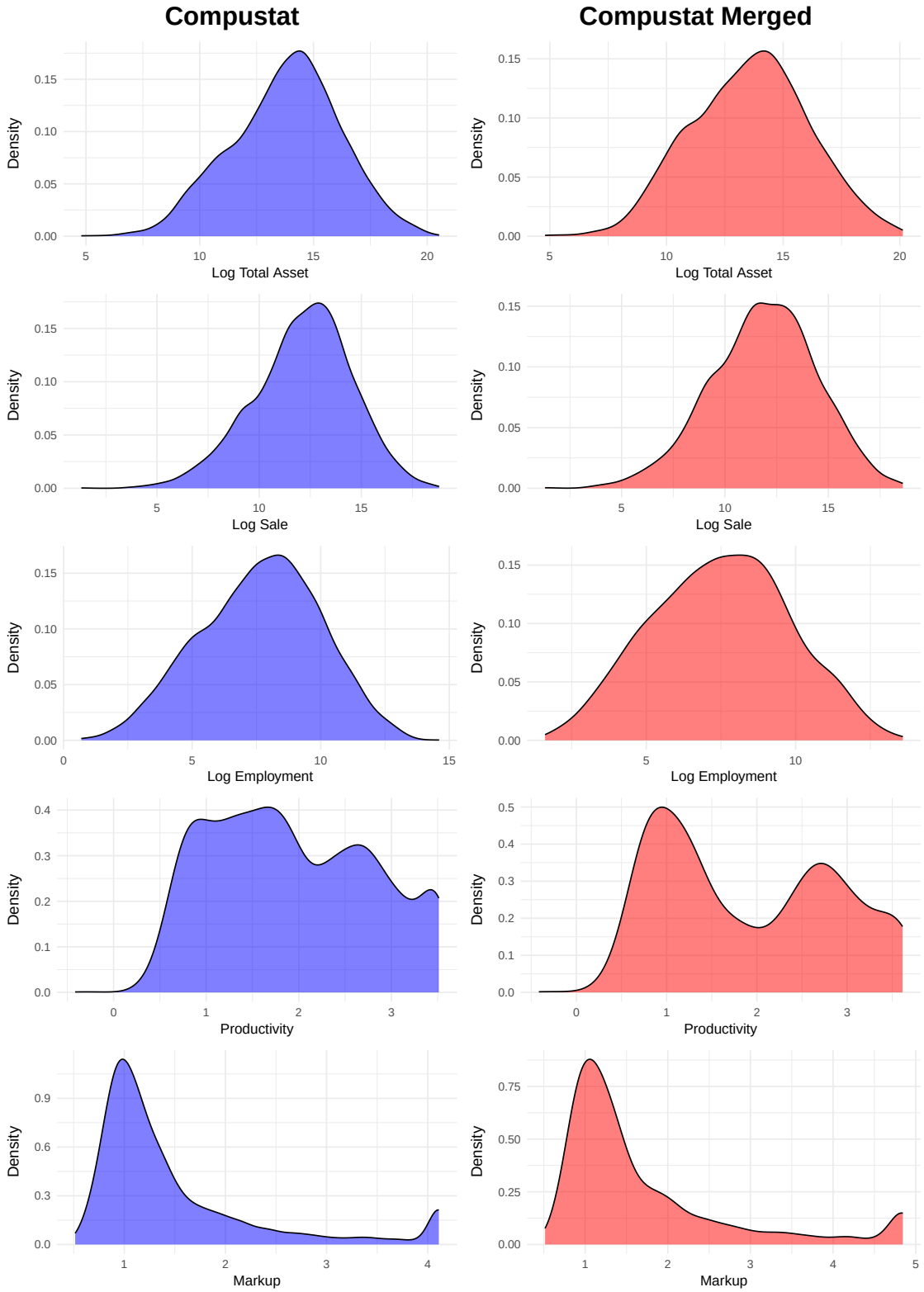

Figure A4. Density Comparison Compustat and Compustat Merged

Note: Both datasets exclude utilities and finance sectors, and all variables are measured for the 2019 cross-section. The Compustat merged dataset further excludes firms with missing or non-positive R&D, SG&A, and segment information.

Table A3. Regression Results: Compustat Dataset

Panel A: *Markup* and *Productivity*

	Pooled OLS		**Two-way FE**	
	(2) *Markup*	(3) *Productivity*	(6) *Markup*	(7) *Productivity*
Production Lines	-0.099^{***}	-0.056^{***}	-0.111^{***}	-0.032^{***}
	(0.007)	(0.010)	(0.008)	(0.004)
Num. Obs.	40,517	40,517	40,517	40,517
Adj. R^2	0.042	0.025	0.111	0.808
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Panel B: *Investment Ratio* and *Fluidity*

	Pooled OLS		**Two-way FE**	
	(1) *Transferable/Embedded*	(4) *Fludity*	(5) *Transferable/Embedded*	(8) *Fludity*
Production Lines	-0.092^{***}	-0.449^{***}	-0.100^{***}	-0.302^{***}
	(0.008)	(0.031)	(0.009)	(0.033)
Num. Obs.	41,071	30,403	41,071	30,403
Adj. R^2	0.085	0.043	0.127	0.169
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Notes: Each column reports coefficients from a separate regression. Standard errors are clustered by *firm id* in parentheses. Significance levels:$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Columns 1–4: Pooled OLS specifications; Columns 5–8: Two-way fixed effects (year and industry).

B. Model Appendix

B.1. Final Good Sector Demand

The final good sector's profit maximization problem is:

$$\max_{y_{jt}} \exp\left(\int_0^1 \ln y_{jt} dj\right) - \int_0^1 p_{jt} y_{jt} dj. \tag{50}$$

The first-order condition yields the inverse demand function for each intermediate good j:

$$p_{jt} = \frac{Y_t}{y_{jt}}. \tag{51}$$

B.2. Intermediate Good Sector Demand

The cost minimization problem for the intermediate sector j is:

$$\min_{y_{sjt}, y_{fjt}} \quad p_{sjt} y_{sjt} + p_{fjt} y_{fjt} \quad s.t. \quad y_{jt} = \left(\chi(e_s) y_{sjt}^{\varepsilon} + y_{fjt}^{\varepsilon}\right)^{\frac{1}{\varepsilon}}. \tag{52}$$

The first-order condition with respect to the superstar firm's output y_{sjt} is:

$$p_{sjt} = \lambda \chi(e_s) y_{jt}^{1-\varepsilon} y_{sjt}^{\varepsilon-1}, \tag{53}$$

where λ is the Lagrange multiplier. Raising both sides to the power of $\frac{\varepsilon}{\varepsilon-1}$ and simplifying allows to solve for the ideal price index p_j for sector j:

$$\lambda = \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}}\right)^{\frac{\varepsilon-1}{\varepsilon}} \equiv p_j. \tag{54}$$

Substituting $\lambda = p_j$ back into the first-order condition yields the inverse demand function faced by the superstar firm:

$$p_{sjt} = p_j \chi(e_s) y_{jt}^{1-\varepsilon} y_{sjt}^{\varepsilon-1}. \tag{55}$$

Finally, substituting the final good producer's demand $y_{jt} = Y_t / p_j$ and solving for y_{sjt} provides demand function:

$$y_{sjt} = p_j^{\frac{\varepsilon}{1-\varepsilon}} \chi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t. \tag{56}$$

B.3. Superstar Firm Maximization Problem: Bertrand Competition

The superstar firm competes à la Bertrand with a continuum of fringe firms. Its profit maximization problem in industry j is:

$$\max_{p_{sjt}} \left(p_{sjt} - MC_{sjt}\right) y_{sjt} \quad s.t. \quad y_{sjt} = p_j^{\frac{\varepsilon}{1-\varepsilon}} \chi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t. \tag{57}$$

Substituting the expression for p_j into the demand function, the objective function can be expanded as:

$$\max_{p_{sjt}} \left[p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} \chi(e_s)^{\frac{1}{1-\varepsilon}} Y_t \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} - MC_{sjt} \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} p_{sjt}^{\frac{1}{\varepsilon-1}} \chi(e_s)^{\frac{1}{1-\varepsilon}} Y_t \right]. \tag{58}$$

After computing the derivative and factoring common terms, this condition can be expressed as:

$$\begin{aligned} \frac{\partial \pi}{\partial p_{sjt}} = Y_t \chi(e_s)^{\frac{1}{1-\varepsilon}} \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \times \Bigg\{ & \left[\frac{\varepsilon}{\varepsilon-1} p_{sjt}^{\frac{1}{\varepsilon-1}} - \text{MC}_{sjt} \frac{1}{\varepsilon-1} p_{sjt}^{\frac{2-\varepsilon}{\varepsilon-1}} \right] \\ & - \left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - \text{MC}_{sjt} p_{sjt}^{\frac{1}{\varepsilon-1}} \right) \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} \frac{\varepsilon}{\varepsilon-1} p_{sjt}^{\frac{1}{\varepsilon-1}} \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \right) \Bigg\} = 0. \end{aligned} \tag{59}$$

$$\begin{aligned} 0 = & \left[\frac{\varepsilon}{\varepsilon-1} p_{sjt}^{\frac{1}{\varepsilon-1}} - \text{MC}_{sjt} \frac{1}{\varepsilon-1} p_{sjt}^{\frac{2-\varepsilon}{\varepsilon-1}} \right] \\ & - \left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - \text{MC}_{sjt} p_{sjt}^{\frac{1}{\varepsilon-1}} \right) \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} \frac{\varepsilon}{\varepsilon-1} p_{sjt}^{\frac{1}{\varepsilon-1}} \left(\chi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \right). \end{aligned} \tag{60}$$

To simplify, multiply both sides by p_{sjt} and substitute the market share definition ϕ_{sjt} from (18), yielding:

$$\left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - \text{MC}_{sjt} p_{sjt}^{\frac{1}{\varepsilon-1}} \right) \left(\phi_{sjt} \frac{\varepsilon}{\varepsilon-1} \right) = \left[\frac{\varepsilon}{\varepsilon-1} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - \text{MC}_{sjt} \frac{1}{\varepsilon-1} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} \right]. \tag{61}$$

First, divide both sides by $p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}}$, then by $\frac{\varepsilon}{\varepsilon-1}$. After rearranging terms, the expression simplifies to:

$$\varepsilon(1-\phi_{sjt}) = \frac{\text{MC}_{sjt}}{p_{sjt}}(1-\varepsilon\phi_{sjt}). \tag{62}$$

Solving for the optimal price p_{sjt} gives:

$$p_{sjt} = \frac{1-\varepsilon\phi_{sjt}}{\varepsilon(1-\phi_{sjt})} \cdot \text{MC}_{sjt}. \tag{63}$$

To determine the optimal labor demand, equating output (6) and demand (56) yields

$$q_{sjt}\,\psi(e_s,n_s)\,l_{sjt} = p_j^{\frac{\varepsilon}{1-\varepsilon}}\chi(e_s)^{\frac{1}{1-\varepsilon}}p_{sjt}^{\frac{1}{\varepsilon-1}}Y_t. \tag{64}$$

Multiplying both sides by p_{sjt} and dividing by w_t gives

$$p_{sjt}\underbrace{\frac{q_{sjt}\,\psi(e_s,n_s)}{w_t}}_{\text{inverse MC}_{sjt}}l_{sjt} = \underbrace{p_j^{\frac{\varepsilon}{1-\varepsilon}}\chi(e_s)^{\frac{1}{1-\varepsilon}}p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}}}_{\phi_{sjt}}\underbrace{\frac{Y_t}{w_t}}_{\omega_t^{-1}}. \tag{65}$$

This expression leads directly to equation (24).

B.4. Superstar Firm Maximization Problem: Cournot Competition

In the à la Cournot setup, a superstar firm and a continuum of fringe firms compete by choosing quantities to sell rather than engaging in price competition as in the à la Bertrand case. Its profit-maximization problem in industry j is

$$\max_{y_{sjt}}\ \pi_{sjt} \ = \ \max_{y_{sjt}}\,(p_{sjt}-MC_{sjt})\,y_{sjt}, \tag{66}$$

$$s.t. \quad p_{sjt} = \chi(e_s)\,y_{jt}^{-\varepsilon}\,y_{sjt}^{\varepsilon-1}\,Y_t, \quad \text{and} \quad y_{jt} = \left(\chi(e_s)\,y_{sjt}^{\varepsilon} + y_{fjt}^{\varepsilon}\right)^{1/\varepsilon}. \tag{67}$$

Differentiate the profit function with respect to y_{sjt}:

$$\frac{d\pi_{sjt}}{dy_{sjt}} = (p_{sjt}-MC_{sjt}) + y_{sjt}\frac{dp_{sjt}}{dy_{sjt}} = 0. \tag{68}$$

Differentiate the inverse demand (67) to obtain

$$\frac{dp_{sjt}}{dy_{sjt}} = p_{sjt}\left(-\varepsilon\frac{1}{y_{jt}}\frac{dy_{jt}}{dy_{sjt}} + \frac{\varepsilon-1}{y_{sjt}}\right). \tag{69}$$

Differentiate y_{jt} with respect to y_{sjt}:

$$\frac{dy_{jt}}{dy_{sjt}} = \chi(e_s)\, y_{sjt}^{\varepsilon-1}\, y_{jt}^{1-\varepsilon}. \tag{70}$$

Substituting (70) and (67) into (68) yields

$$(p_{sjt} - MC_{sjt}) + y_{sjt}\, p_{sjt}\left(-\varepsilon\chi(e_s)\, y_{sjt}^{\varepsilon-1} y_{jt}^{-\varepsilon} + \frac{\varepsilon-1}{y_{sjt}}\right) = 0. \tag{71}$$

Using the market-share definition

$$\frac{p_{sjt}\, y_{sjt}}{Y_t} = \chi(e_s)\, y_{sjt}^{\varepsilon}\, y_{jt}^{-\varepsilon},$$

and dividing both sides by p_{sjt}, rearrangement gives the inverse markup condition

$$\frac{MC_{sjt}}{p_{sjt}} = \varepsilon(1-\phi_{sjt}). \tag{72}$$

The relative price ratio between fringe and superstar firms is therefore

$$\frac{p_{fjt}}{p_{sjt}} = \frac{1-\phi_{sjt}}{\phi_{sjt}}\,\frac{MC_{fjt}}{MC_{sjt}}. \tag{73}$$

Using the inverse demand expressions leads to

$$\chi(e_s)\left(\frac{y_{fjt}}{y_{sjt}}\right)^{\varepsilon-1} = \frac{1-\phi_{sjt}}{\phi_{sjt}}\,\frac{MC_{fjt}}{MC_{sjt}}. \tag{74}$$

Substituting the marginal-cost expressions for the fringe and superstar firms yields

$$\left(\frac{y_{fjt}}{y_{sjt}}\right)^{\varepsilon-1} = \frac{1}{\chi(e_s)}\,\frac{1-\phi_{sjt}}{\phi_{sjt}}\,\frac{1}{\lambda^{m_j}\,\psi(e_s, n_s)}. \tag{75}$$

Equation (75) shows that the relative output depends on the market share, the quality gap, the embedded intangible level, and the firm's production-line parameter. Rearranging the market-share definition gives

$$\chi(e_s)\, y_{sjt}^{\varepsilon}\, y_{jt}^{-\varepsilon} = \frac{\chi(e_s)\, y_{sjt}^{\varepsilon}}{\chi(e_s)\, y_{sjt}^{\varepsilon} + y_{fjt}^{\varepsilon}} = \frac{1}{1 + \dfrac{1}{\chi(e_s)} \left(\dfrac{y_{fjt}}{y_{sjt}}\right)^{\varepsilon}}, \tag{76}$$

which shows that market share depends on the output gap and the embedded intangible level. Therefore, the output gap depends on the quality gap, the embedded intangible level, and the number of production lines associated with superstar s.

B.5. Aggregate Output and Growth Rate

Using the superstar (6) and fringe firm (7) output equations into (3) gives

$$Y_t = \exp\left(\int_0^1 \ln\left[(\xi e_{st})^{\beta}\Big(q_{sjt}\,\frac{\big((1-\xi)e_{st}\big)^{\alpha}}{\gamma n_{st}^{\alpha_s}}\, l_{sjt}\Big)^{\varepsilon} + \big(q_{fjt}\, l_{fst}\big)^{\varepsilon}\right]^{1/\varepsilon} dj\right)$$

$$Y_t = \exp\left(\int_0^1 \ln\left[\left(q_{sjt}^{\varepsilon}\left((\xi e_{st})^{\beta}\Big(\frac{\big((1-\xi)e_{st}\big)^{\alpha}}{\gamma n_{st}^{\alpha_s}}\, l_{sjt}\Big)^{\varepsilon} + \big(\lambda^{-m_{jt}}\, l_{fst}\big)^{\varepsilon}\right)\right)\right]^{1/\varepsilon} dj\right)$$

$$= \underbrace{\exp\left(\int_0^1 \ln q_{sjt} dj\right)}_{Q_t} \exp\left(\int_0^1 \ln\left[((\xi e_{st})^{\beta}\left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}}\, l_{sjt}\right)^{\varepsilon} + \Big(\lambda^{-m_{jt}}\, l_{fst}\Big)^{\varepsilon}\right]^{\frac{1}{\varepsilon}} dj\right), \tag{77}$$

and , along with the labor demands in (24) and (25), and factoring out ω_t^{-1}, aggregate output can be expressed as

$$Y_t = Q_t\, \omega_t^{-1} \exp\left(\int_0^1 \ln\left[(\xi e_{st})^{\beta}\left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}}\frac{\phi_{sjt}}{\sigma_{sjt}}\right)^{\varepsilon} + \Big(\lambda^{-m_{jt}}(1-\phi_{sjt})\Big)^{\varepsilon}\right]^{\frac{1}{\varepsilon}} dj\right). \tag{78}$$

Because everything in the integrand depends only on the gaps (m, k, n), the expression can be written in discrete state space as

$$Y_t = Q_t\,\omega_t^{-1}\,\exp\left(\sum_{m=1}^{\bar{m}}\sum_{k=1}^{\bar{k}}\sum_{n=1}^{\bar{n}} \underbrace{\ln\left[\left(\xi\theta^k\right)^\beta\left(\frac{((1-\xi)\theta^{kt})^\alpha}{\gamma n^{\alpha_s}}\frac{\phi_t(m,k,n)}{\sigma_t(m,k,n)}\right)^\varepsilon + \left(\lambda^{-mt}(1-\phi_t(m,k,n))\right)^\varepsilon\right]^{\frac{1}{\varepsilon}}}_{\equiv R_t(m,k,n)} \mu_t(m,k,n)\right). \tag{79}$$

$$\begin{aligned}\ln Y_{t+\Delta t} - \ln Y_t = &\left(\ln Q_{t+\Delta t} - \ln Q_t\right) + \ln\omega_t - \ln\omega_{t+\Delta t}\\ &+ \sum_{m=1}^{\bar{m}}\sum_{k=1}^{\bar{k}}\sum_{n=1}^{\bar{n}}\Big(R_{t+\Delta t}(m,k,n) - R_t(m,k,n)\Big)\Big(\mu_{t+\Delta t}(m,k,n) - \mu_t(m,k,n)\Big) + o(\Delta t).\end{aligned} \tag{80}$$

where

$$\begin{aligned}\ln Q_{t+\Delta t} - \ln Q_t = \ln\lambda &\left[\sum_{m=1}^{\bar{m}}\sum_{k=1}^{\bar{k}}\sum_{n=1}^{\bar{n}}\Big(z_t^{Int}(m,k,n) + p_{s\geq s'}^{\text{Ex}}\, z_t^{Ex}(m,k,n) + Z_t^f(m,k,n)\Big)\mu_t(m,k,n)\right]\Delta t\\ &+ o(\Delta t).\end{aligned} \tag{81}$$

Dividing by Δt and taking the limit $\Delta t \to 0$, the growth rate of the economy is

$$g_t = -g_{\omega,t} + g_{Q,t} + g_{R,t}. \tag{82}$$

In the steady state, the distribution $\mu_t(m, k, n)$ is constant, implying that R_t is constant. Wages grow at the same rate as output, so the real wage remains constant. Therefore, in steady state the growth rate of the economy is determined solely by quality improvements:

$$g = g_Q = \ln\lambda\left[\sum_{m=1}^{\bar{m}}\sum_{k=1}^{\bar{k}}\sum_{n=1}^{\bar{n}}\Big(z^{\text{Int}}(m,k,n) + p_{s\geq s'}^{\text{Ex}}\, z^{\text{Ex}}(m,k,n) + Z^f(m,k,n)\Big)\mu(m,k,n)\right]. \tag{83}$$

B.6. Decomposition of Output

Starting from (78) and factoring out the term $(\xi e_{st})^{\beta}\left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma\, n_{st}^{\alpha_s}}\right)$, aggregate output can be written as

$$Y_t = Q_t\, \omega_t^{-1}\, \exp\left(\int_0^1 \ln\left[\left(\frac{\phi_{sjt}}{\sigma_{sjt}}\right)^{\varepsilon} + \frac{1}{(\xi e_{st})^{\beta}}\left(\frac{\gamma\, n_{st}^{\alpha_s}}{((1-\xi)e_{st})^{\alpha}}\, \lambda^{-m_{jt}}(1-\phi_{sjt})\right)^{\varepsilon}\right]^{1/\varepsilon} dj\right). \tag{84}$$

Define the multiplicative factor that collects the factored-out terms as

$$E_t \;=\; \exp\left(\int_0^1 \ln\left[(\xi e_{st})^{\beta}\left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma\, n_{st}^{\alpha_s}}\right)^{\varepsilon}\right]^{1/\varepsilon} dj\right). \tag{85}$$

Next multiply and divide the integrand by the linear weight $\dfrac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})$. After this algebraic step I obtain a decomposition that isolates a simple mean term and a residual term:

$$\begin{aligned} Y_t = Q_t\, E_t\, \omega_t^{-1}\, &\exp\left(\int_0^1 \ln\left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})\right] dj\right) \\ &\times \exp\left(\int_0^1 \ln\left[\frac{\left(\frac{1}{\gamma n_{st}^{\alpha_s}}\frac{\phi_{sjt}}{\sigma_{sjt}}\right)^{\varepsilon} + \frac{1}{(\xi e_{st})^{\beta}}\left(\frac{1}{((1-\xi)e_{st})^{\alpha}}\lambda^{-m_{jt}}(1-\phi_{sjt})\right)^{\varepsilon}}{\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})}\right]^{1/\varepsilon} dj\right). \end{aligned} \tag{86}$$

Finally, using (37) and defining the multiplicative mean term

$$M_t \;=\; \frac{\exp\left(\int_0^1 \ln\left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})\right] dj\right)}{\int_0^1 \left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})\right] dj}, \tag{87}$$

the output decomposition can be written as

$$Y_t \;=\; Q_t \times E_t \times M_t \times S_t,$$

where

$$S_t = \exp\left(\int_0^1 \ln\left[\frac{\left(\frac{1}{\gamma\, n_{st}^{\alpha_s}}\frac{\phi_{sjt}}{\sigma_{sjt}}\right)^{\varepsilon} + \frac{1}{(\xi e_{st})^{\beta}}\left(\frac{1}{((1-\xi)e_{st})^{\alpha}}\,\lambda^{-m_{jt}}(1-\phi_{sjt})\right)^{\varepsilon}}{\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})}\right]^{1/\varepsilon} dj\right). \quad (88)$$

B.7. Consumption Equivalence Welfare Measure

On the balanced growth path, consumption grows at rate g, so that $C(t) = C_0 \exp(gt)$.[35] Defining welfare Ω as the present value of lifetime utility from consumption yields:

$$\Omega = \int_0^{\infty} e^{-\rho t} \ln(C(t))\mathrm{d}t \quad (90)$$

$$= \ln(C_0)\int_0^{\infty} e^{-\rho t}\mathrm{d}t + g\int_0^{\infty} te^{-\rho t}\mathrm{d}t. \quad (91)$$

Solving these integrals gives:

$$\Omega = \frac{\ln(C_0)}{\rho} + \frac{g}{\rho^2} \quad (92)$$

$$= \frac{1}{\rho}\left(\ln C_0 + \frac{g}{\rho}\right). \quad (93)$$

Equivalent Welfare Changes Between Economies To compare welfare between two economies—a calibrated benchmark economy (Cal) and a taxed economy (Tax) on their respective balanced growth paths—I compute the percentage change δ in lifetime consumption that would make households indifferent between the two. The required compensation δ satisfies:

$$\Omega^{\text{Tax}} = \frac{1}{\rho}\left(\ln\left(C_0^{\text{Cal}}(1+\delta)\right) + \frac{g^{\text{Cal}}}{\rho}\right). \quad (94)$$

[35] The C_0 consumption level is given by:

$$C_0 = Y_0 - \int_0^1 \left(I_0^{\text{Int}} + I_0^{\text{Emb}} + I_0^{\text{Ex}} + I_0^{f}\right) dj + G_0. \quad (89)$$

In this equation, the subscript 0 represents calibrated optimum values on the balanced growth path, and G is the lump-sum transfer of government taxes.

Solving equation (94) for δ:

$$\frac{\ln C_0^{\text{Tax}}}{\rho} + \frac{g^{\text{Tax}}}{\rho^2} = \frac{\ln[C_0^{\text{Cal}}(1+\delta)]}{\rho} + \frac{g^{\text{Cal}}}{\rho^2} \tag{95}$$

$$\ln\left(\frac{C_0^{\text{Tax}}}{C_0^{\text{Cal}}}\right) + \frac{g^{\text{Tax}} - g^{\text{Cal}}}{\rho} = \ln(1+\delta) \tag{96}$$

$$\delta = \frac{C_0^{\text{Tax}}}{C_0^{\text{Cal}}} \exp\left(\frac{g^{\text{Tax}} - g^{\text{Cal}}}{\rho}\right) - 1. \tag{97}$$

If $\delta > 0$: households require compensation to remain in the benchmark economy (Cal).

If $\delta < 0$: households would pay to move to the taxed economy (Tax).

C. Numerical Appendix

C.1. Additional Numerical Results

Table C1. Sensitivity Matrix

Parameter	Markup	Growth Rate	Investment Ratio	Innovation Rate
ε	-1.052%	-2.113%	-25.815%	-3.763%
α	-0.010%	-0.324%	-3.736%	-0.521%
α_s	0.000%	-0.027%	-0.043%	-0.062%
β	-0.002%	-0.064%	-0.862%	-0.106%
γ	-0.013%	-0.398%	-4.342%	-0.639%
ξ	-0.004%	-0.129%	-1.484%	-0.210%

Note: Each row reports the percentage change in variables resulting from a 1% change in the parameter value.

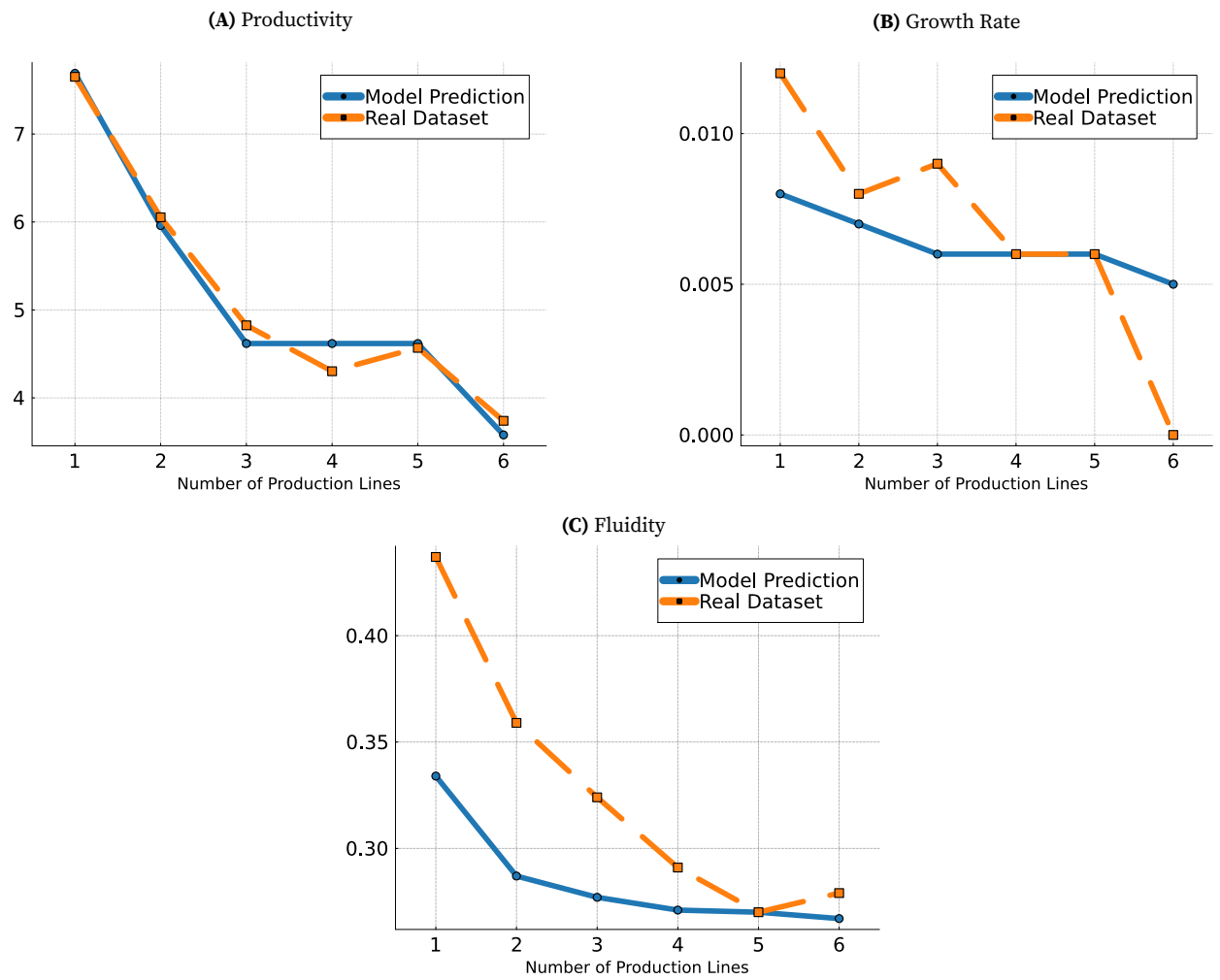

Figure C1. Untargeted Moments: Productivity, Fluidity and Growth Rate by Production Lines

Note: The orange line represents the dataset values, while the blue line shows the model simulation results along the balanced growth path. The horizontal axis corresponds to the production line dimension.

C.2. Solution Algorithm

This algorithm computes the balanced growth path with a three dimensional state space (m, k, n). The solution involves finding the value functions $v_s(m, k, n)$ and $v_f(m, k, n)$, the innovation rates $z^{\text{Int}}, z^{\text{Emb}}, z^{\text{Ex}}, z^f$, and the stationary distribution $\mu(m, k, n)$ that jointly satisfy the model's equilibrium conditions.

BGP Equilibrium Solution:

1. **Compute static values:** Calculate static market shares and profit values using equa-

tions (22) and (18).

2. **Initialization:** Initialize the value functions $v_s(m,k,n)$ and $v_f(m,k,n)$, and the stationary distribution $\mu(m,k,n)$.
3. **Step 1: Solve HJB Equations (Backward Iteration)**
   (a) Set $v^{\text{old}}(m,k,n)$.
   (b) Repeat until $\max|v^{\text{new}} - v^{\text{old}}| <$ tolerance:
      - Compute policy functions $x(m,k,n)$ from the FOCs using v^{old}.
      - Solve the discretized HJB equations for $v^{\text{new}}(m,k,n)$.
      - Update $v^{\text{old}} \leftarrow v^{\text{new}}$.
4. **Step 2: Solve the Kolmogorov Forward Equation (KFE)**
   (a) Set $\mu^{\text{old}}(m,k,n)$.
   (b) Repeat until $\max|\mu^{\text{new}} - \mu^{\text{old}}| <$ tolerance:
      - Solve the discretized KFE for $\mu^{\text{new}}(m,k,n)$ using the policy functions $z^{\text{Int}}, z^{\text{Emb}}, z^{\text{Ex}}, z^{f}$.
      - Update $\mu^{\text{old}} \leftarrow \mu^{\text{new}}$.
5. **Step 3: Repeat Steps Until Value Functions and Distribution Converge**

Finally, to determine the optimal parameter values, search over the parameter space to minimize the objective function,

$$\text{Minimize}(z) = \sum_{z=1}^{Z} \frac{\big|\text{model}(z) - \text{data}(z)\big|}{\frac{1}{2}\big|\text{model}(z)\big| + \frac{1}{2}\big|\text{data}(z)\big|}.$$